

计算机视觉导论



Lec.01: Introduction

2022.09.12

Slides adapted from Noah Snavely, Shree Nayar, T. C. Chang, Lingqi Yan, Brett Bernstein

个人简介

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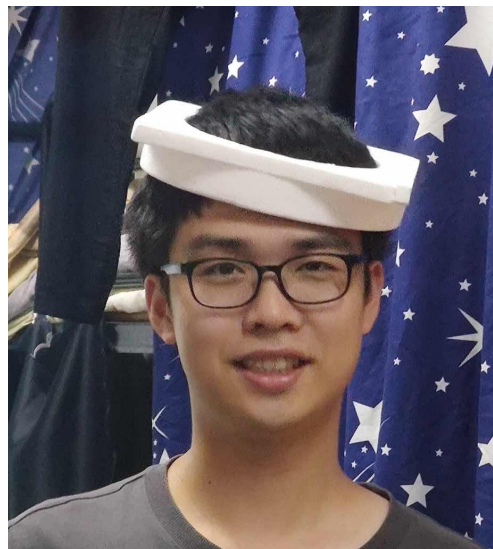
负责图像特征匹配、SfM、
计算摄影学...



林浩通

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负责相机成像，图像信息
处理...



皮怀瑾

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负责深度学习，图像识别...



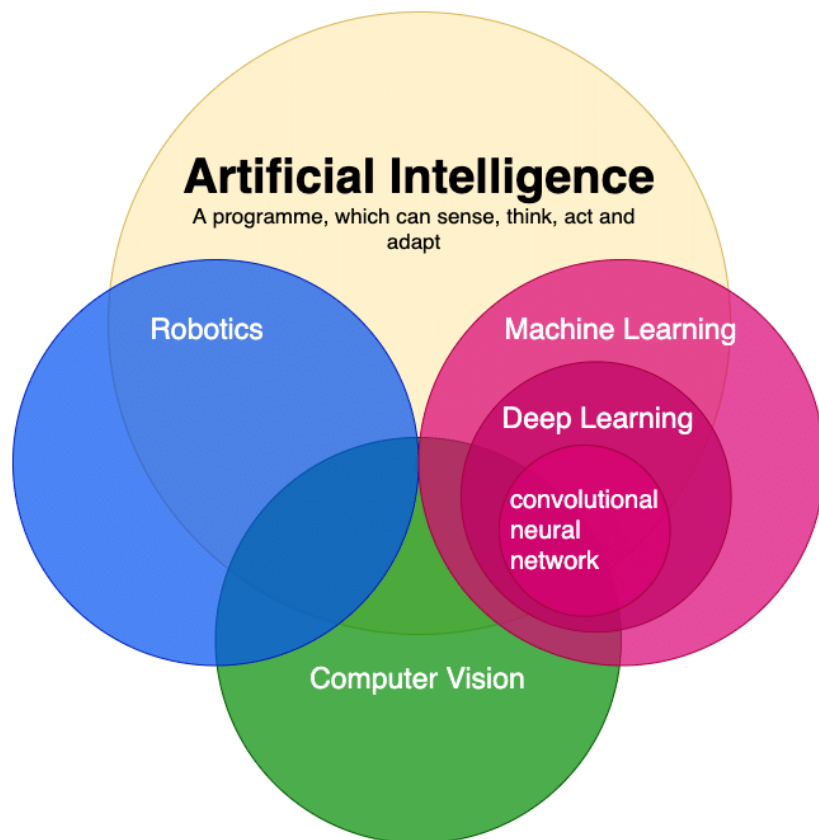
贺星毅

xingyihe@zju.edu.cn

负责深度估计，三维重建，
基于图像的渲染...

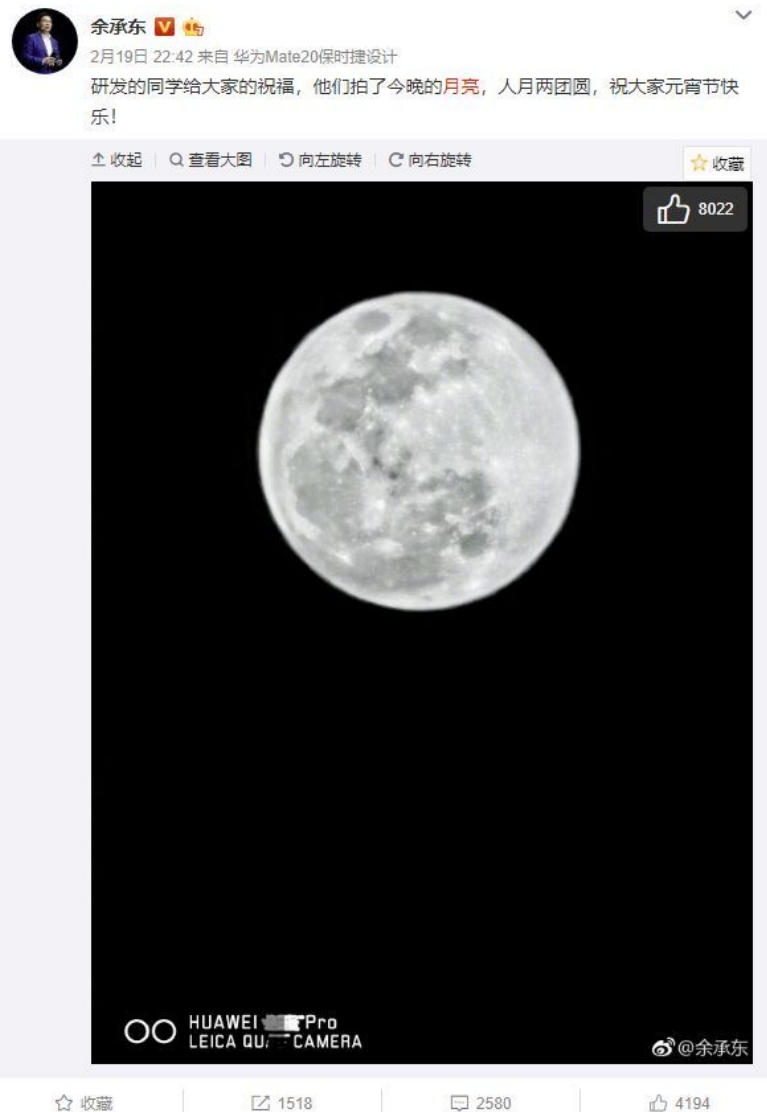
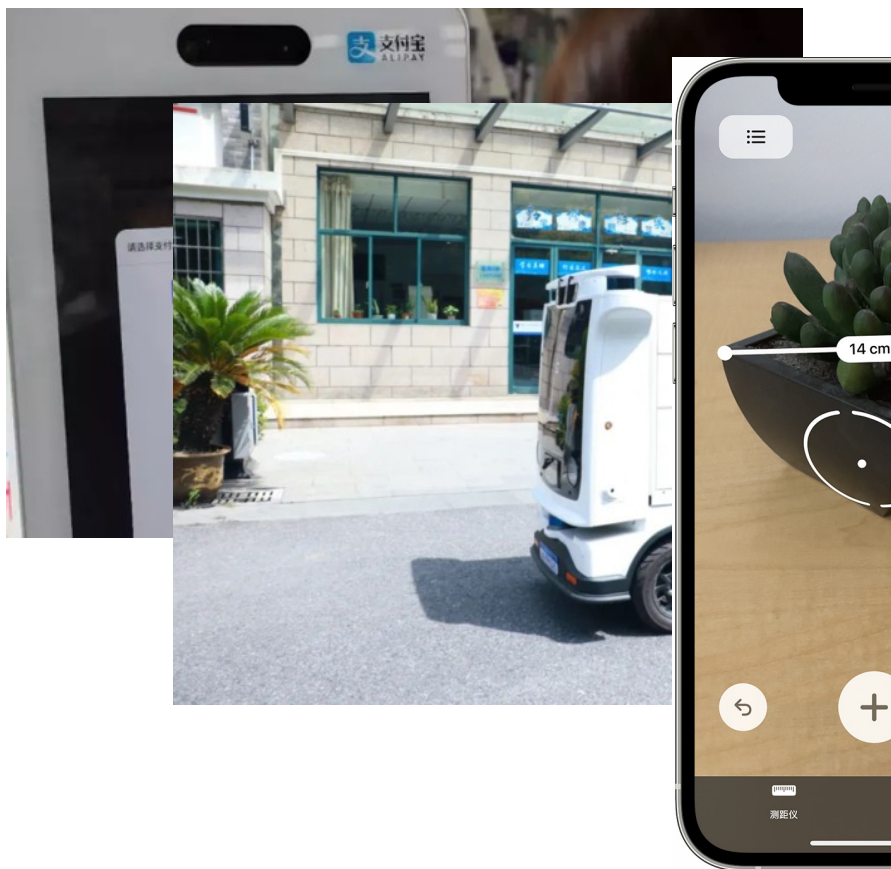
为什么上这门课

- 计算机视觉是人工智能的重要组成部分
- 与很多学科关系紧密



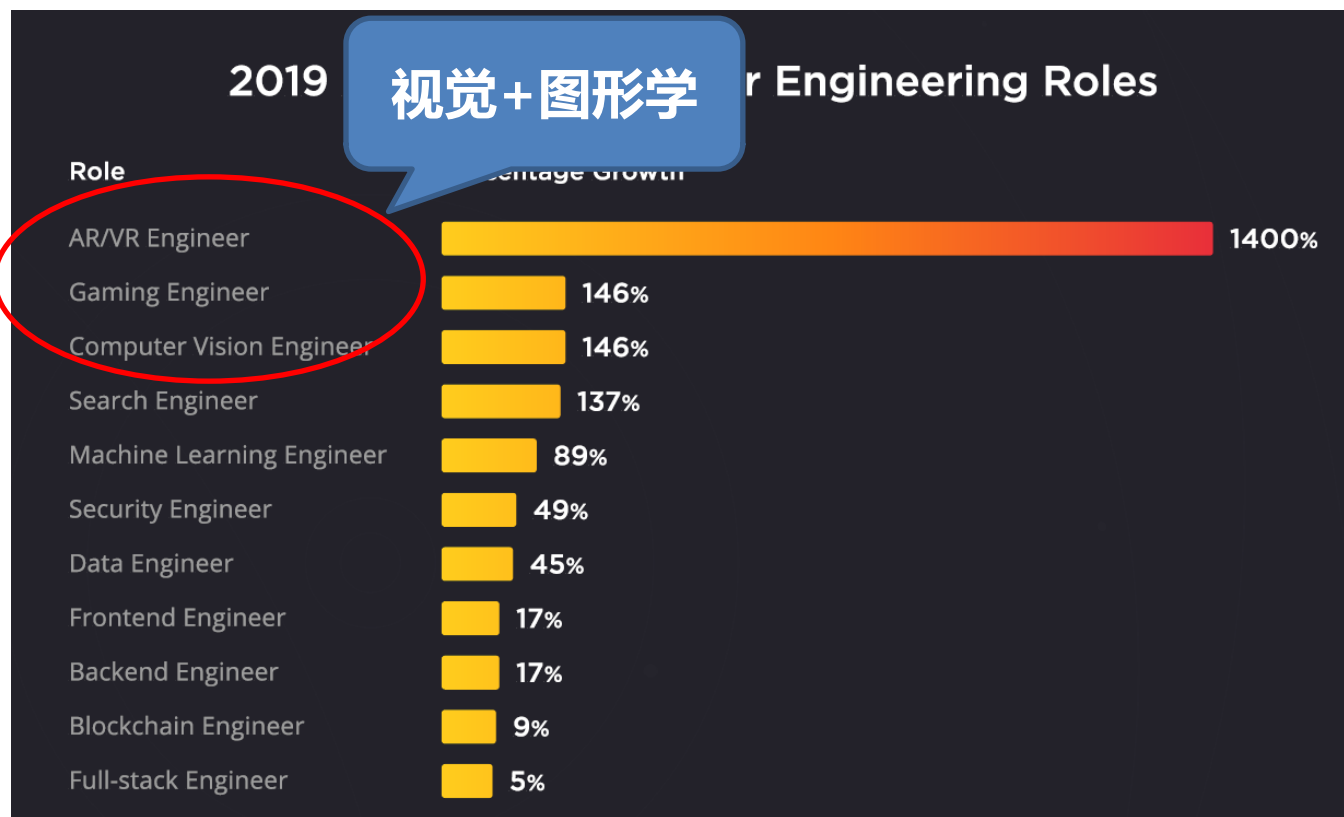
为什么上这门课

- 计算机视觉生动有趣、贴近生活



为什么上这门课

- 计算机视觉应用广泛、工业界需求火热



<https://hired.com/state-of-software-engineers>

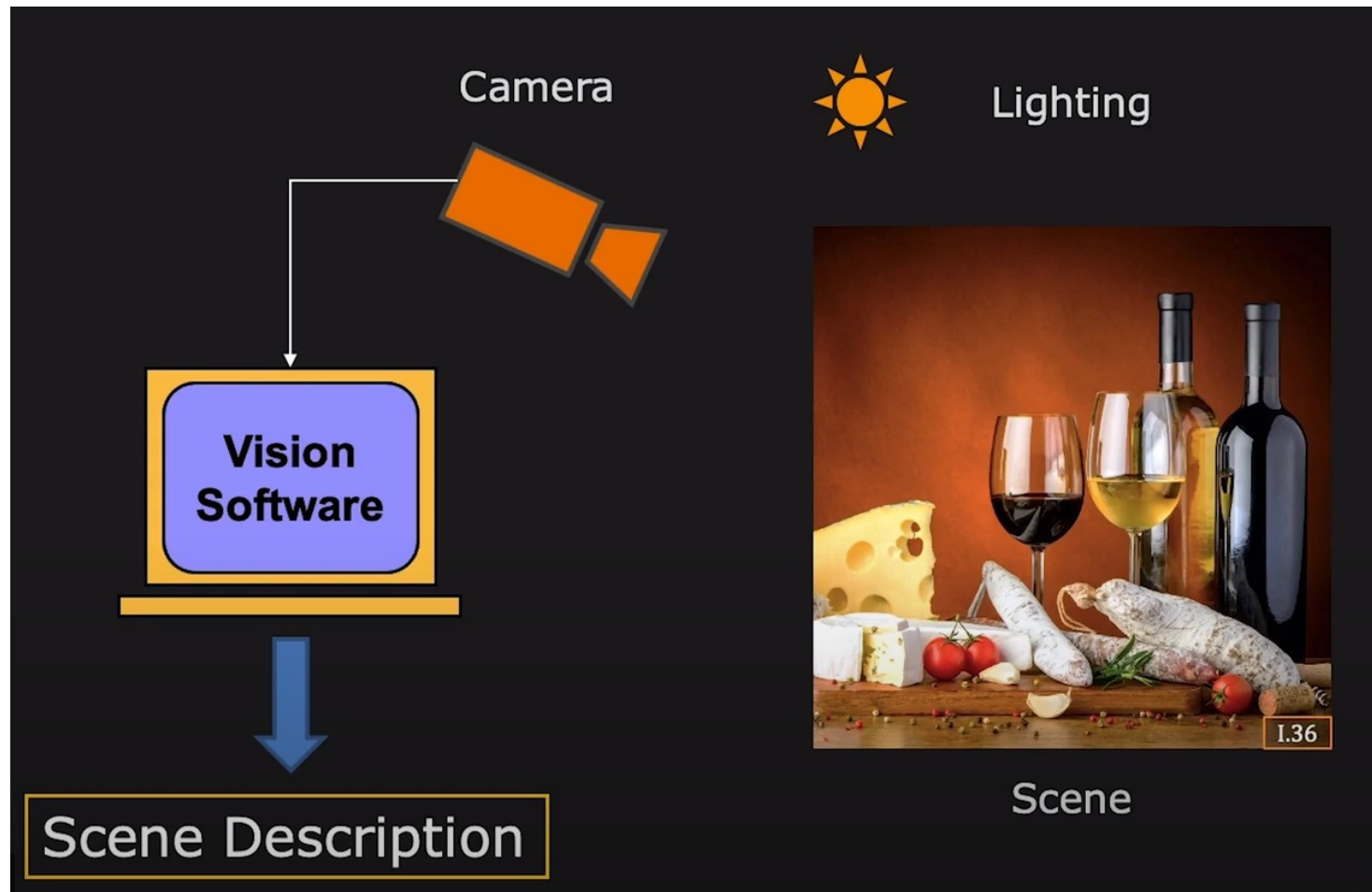
希望你

- 以掌握知识与技能为目标
- 不懂就问
 - 包括英文单词和术语
 - 上课可随时打断老师
 - 程序实现细节问TA
- 千万不要抄袭作业

Today

- **What is Computer Vision?**
- What is Vision Used For?
- Course Overview.
- Review of Linear Algebra.

A standard computer vision system



Credit: <https://fpcv.cs.columbia.edu/>

What really is vision?



Compute properties of the world.

- 3D structure?
- Names of object?
- What happened?

What really is vision?



Compute properties of the world.

- 3D structure?
- Names of object?
- What happened?

Synthesize a color image?

Computer vision tasks

3D reconstruction (三维重建)

- 3D reconstruction, localization, SLAM, ...

Image understanding (图像理解)

- Image recognition, object detection, image segmentation, action recognition, deep learning ...

Image synthesis (图像合成)

- Colorization, super-resolution, deblurring, HDR, panorama, free-viewpoint, GANs, ...

Why is computer vision hard?

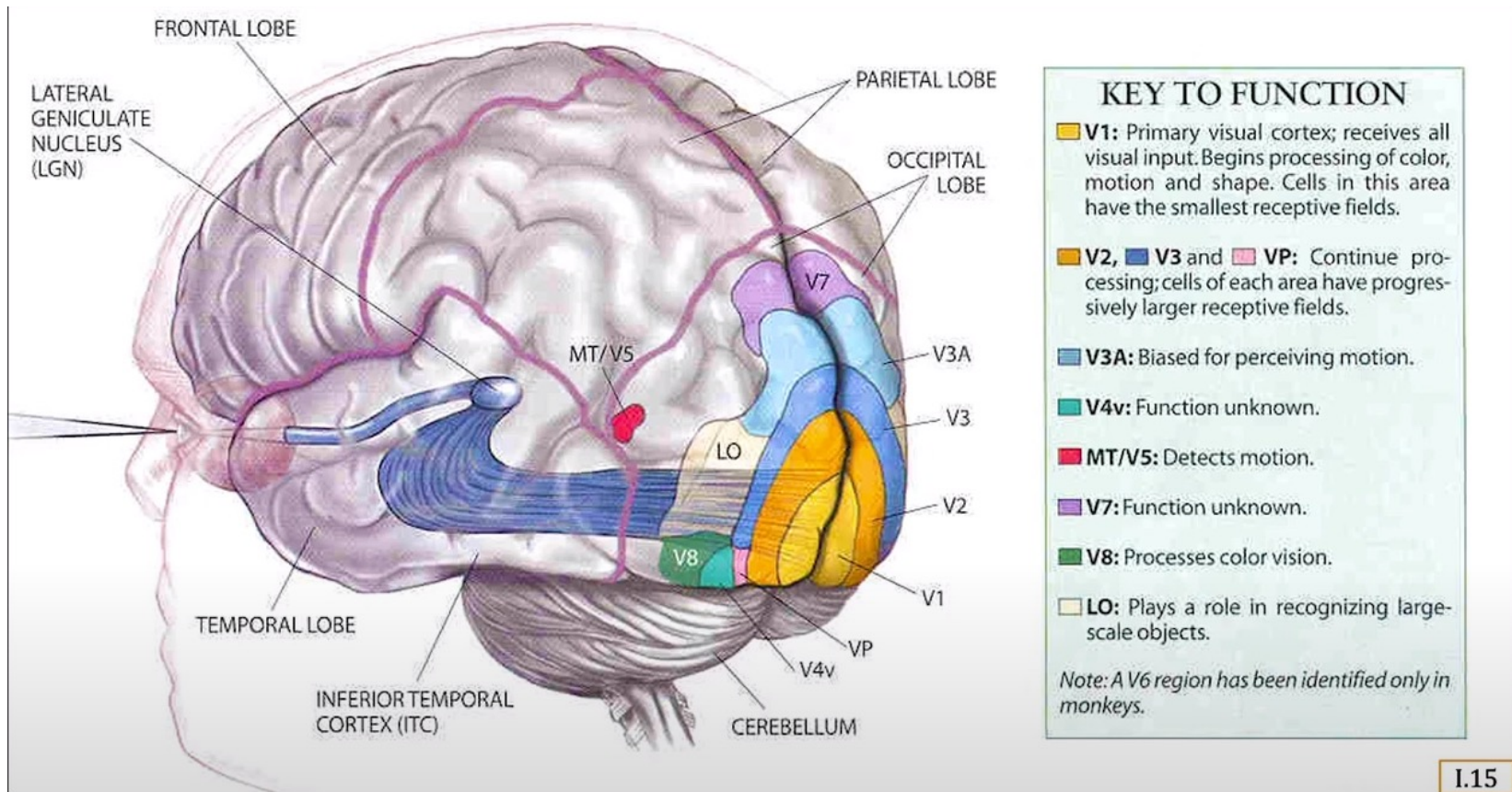


What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What computers see

Why is computer vision hard?



Credit: <https://fpcv.cs.columbia.edu/>

Vision is easy for us. But, we don't know how we do it.

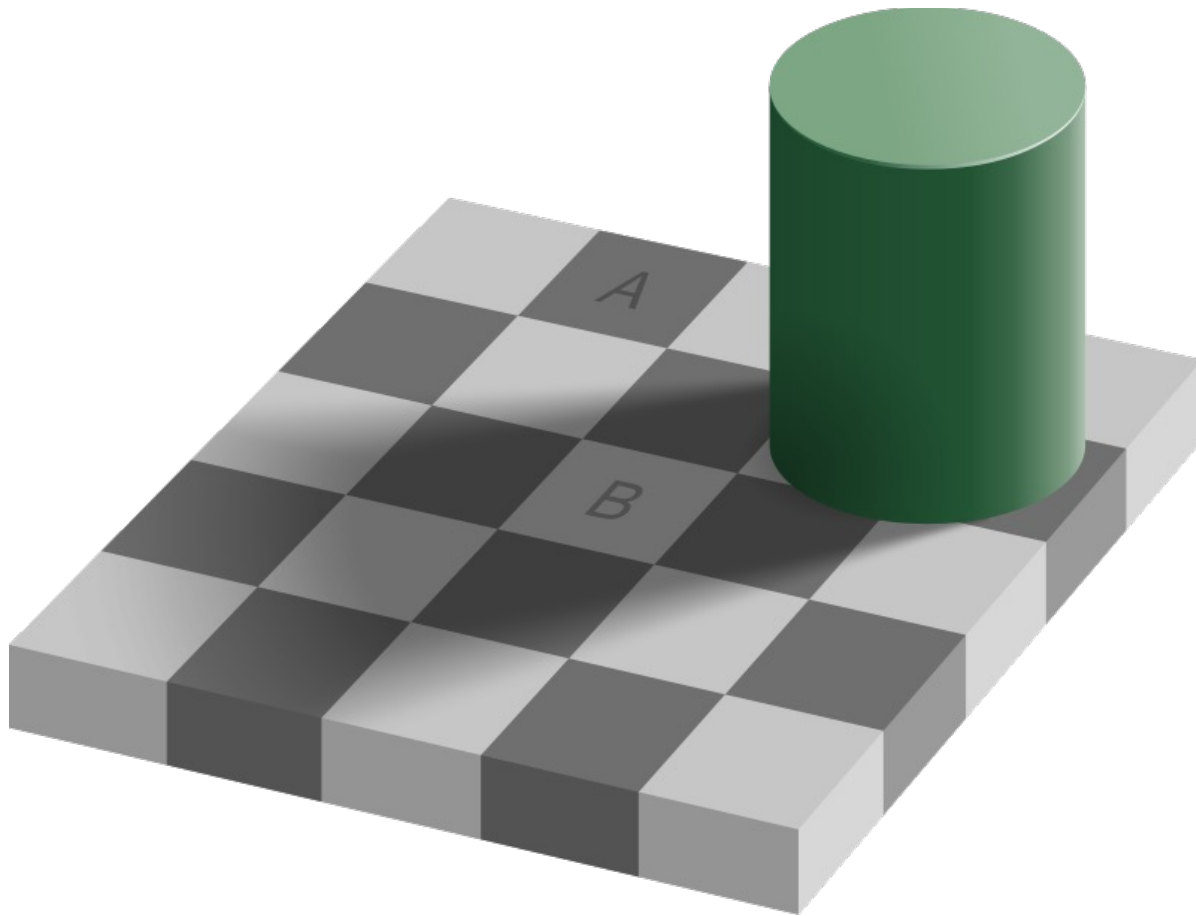
Can computers match human perception?



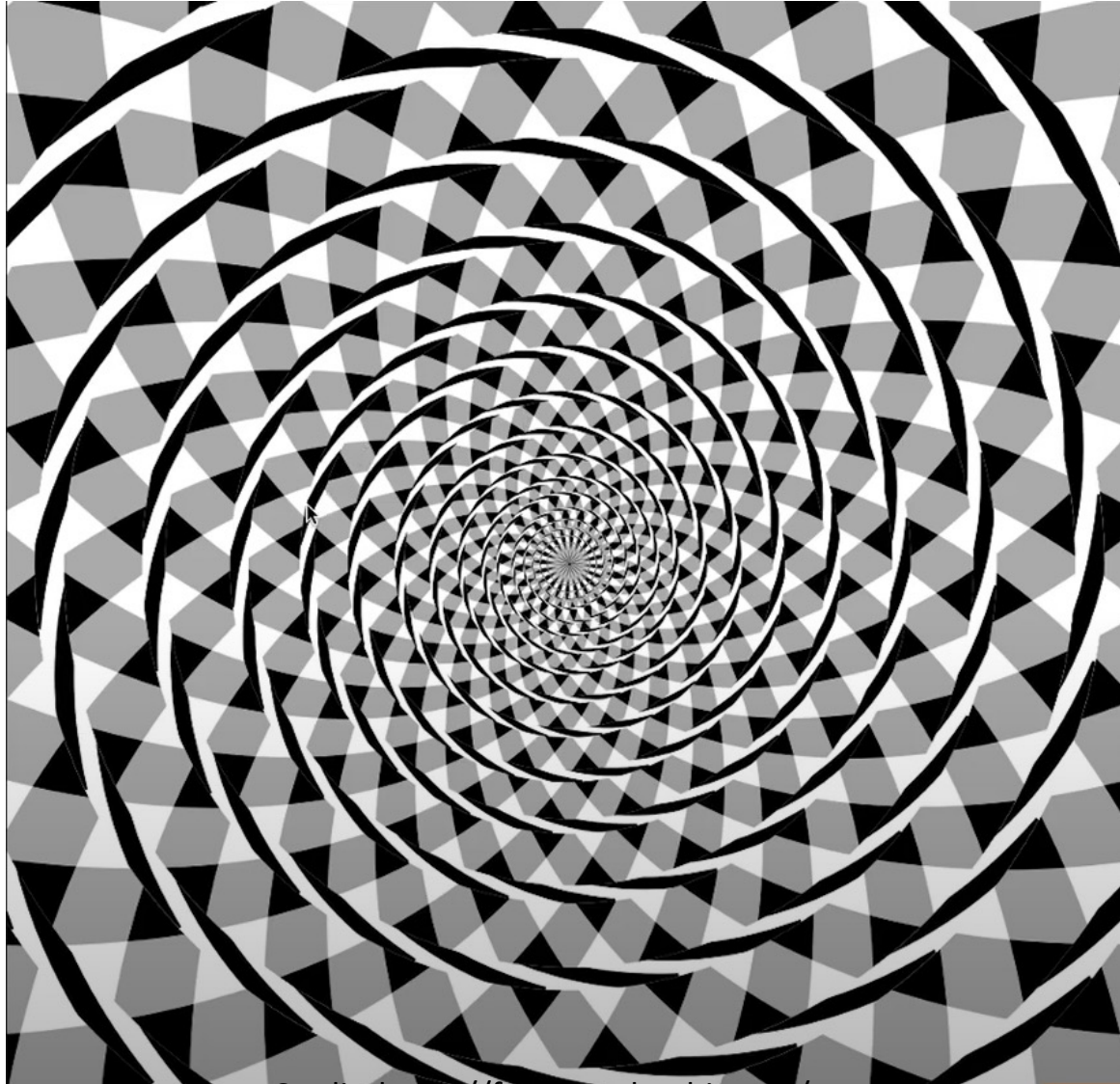
- Yes and no
 - computers can be better at computing.
 - humans are better at understanding (intelligence).

Credit: <https://www.cs.cornell.edu/courses/cs5670/2021sp/>

Human perception has its Shortcomings.

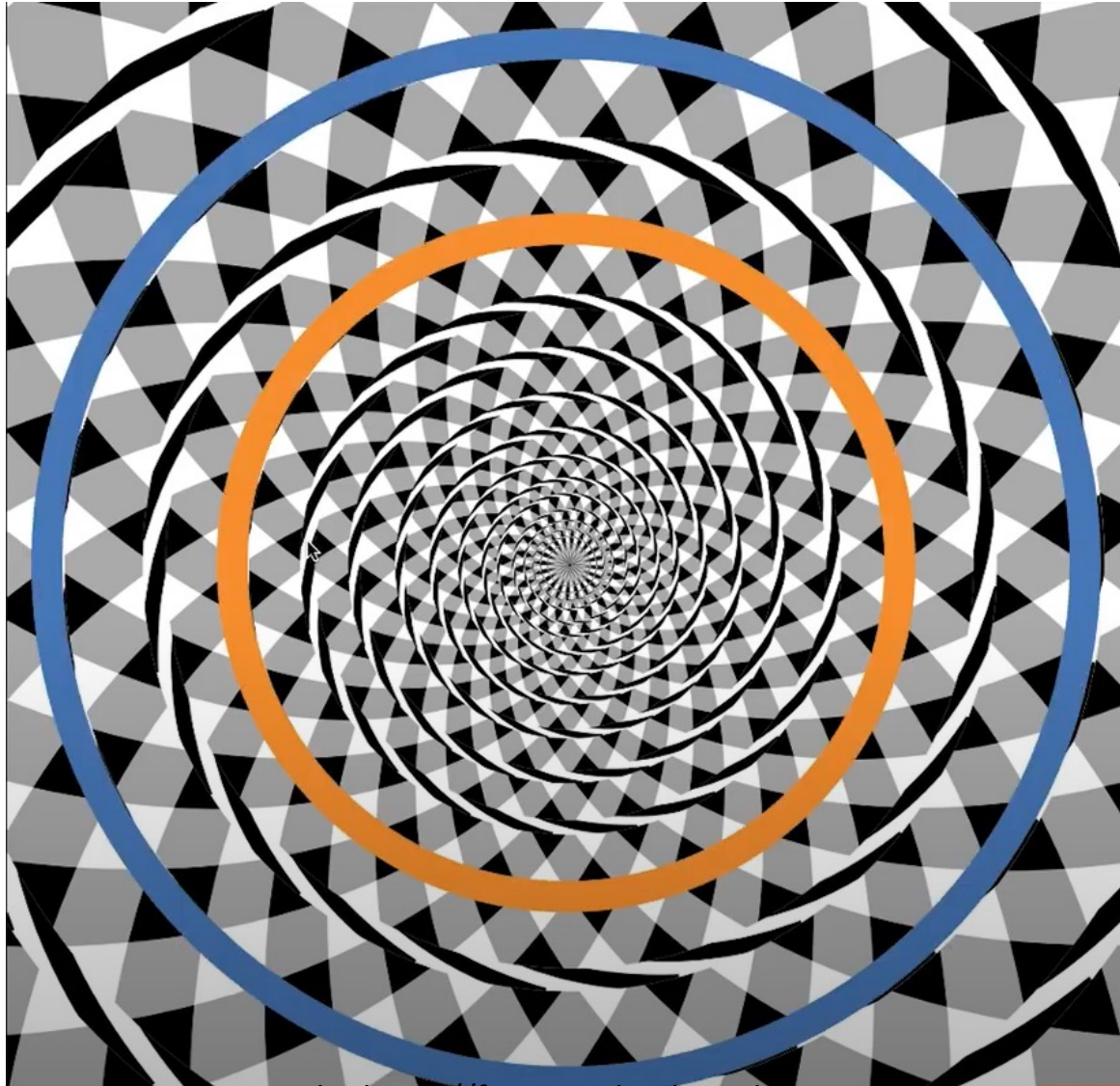


Human perception has its Shortcomings.



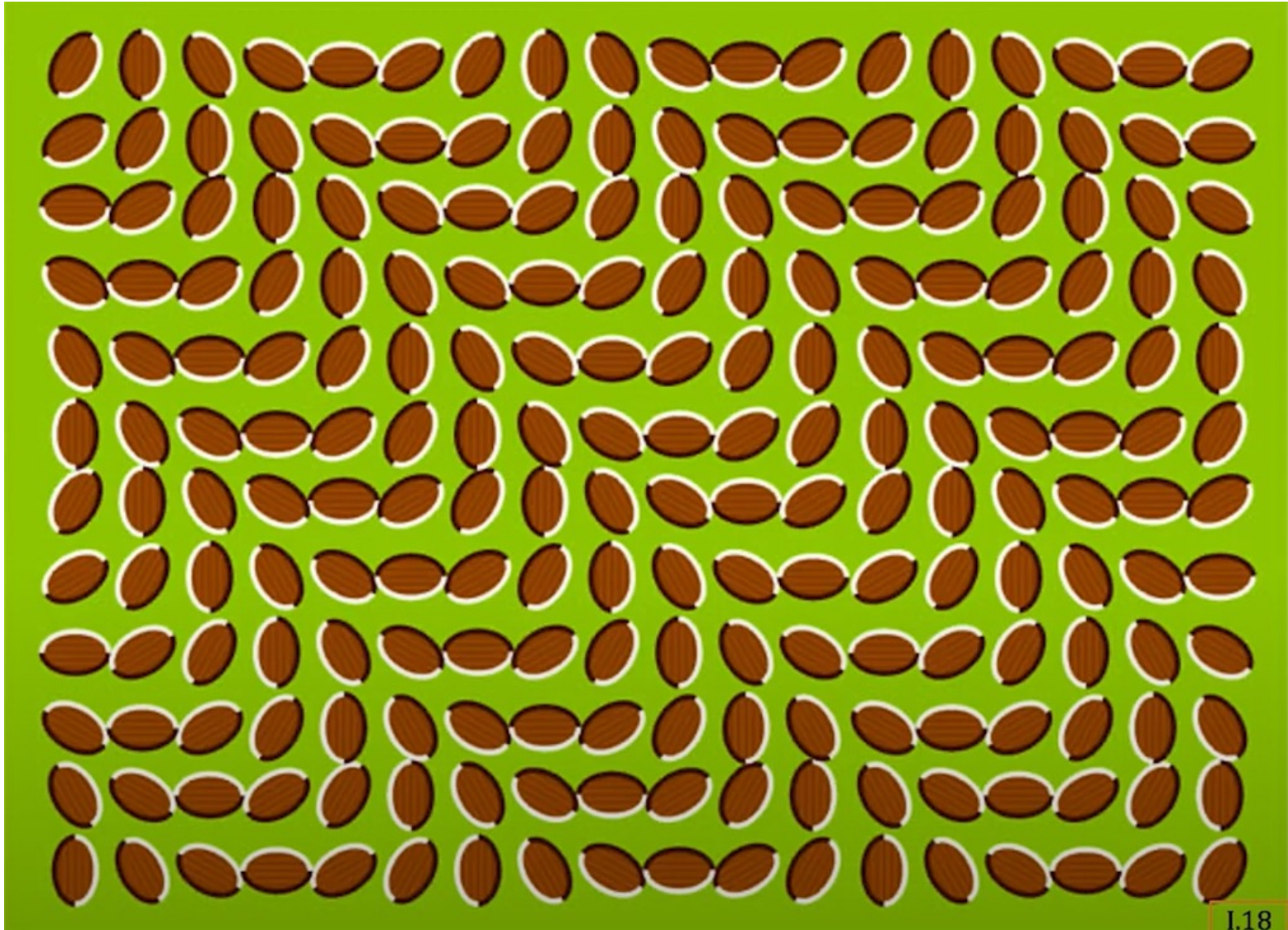
Credit: <https://fpcv.cs.columbia.edu/>

Human perception has its Shortcomings.



Credit: <https://fpcv.cs.columbia.edu/>

Human perception has its Shortcomings.

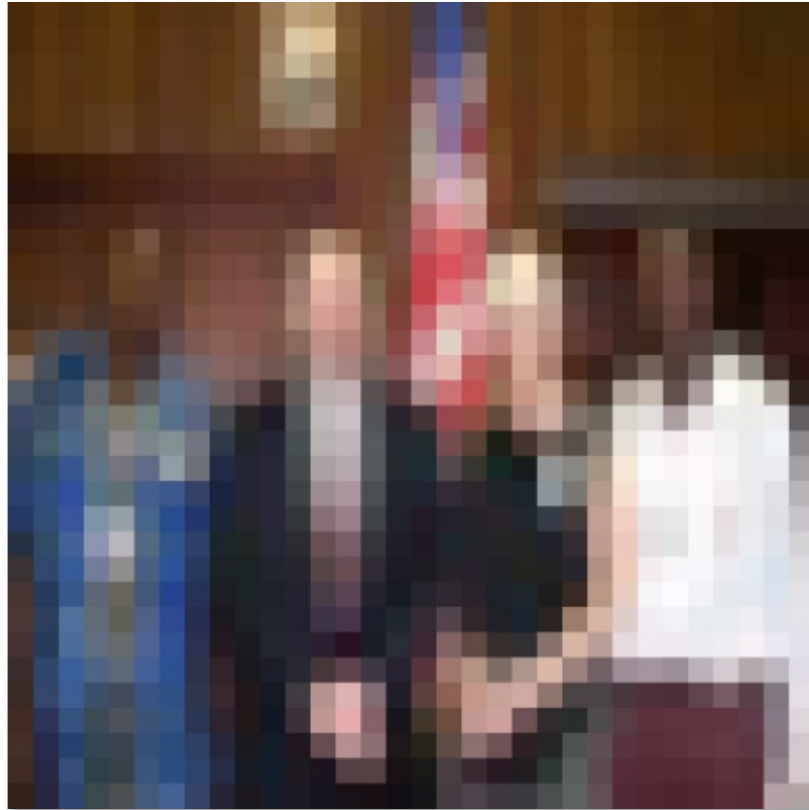


Human perception has its Shortcomings.



Spin lady

But humans can tell a lot about a scene
from a little information...



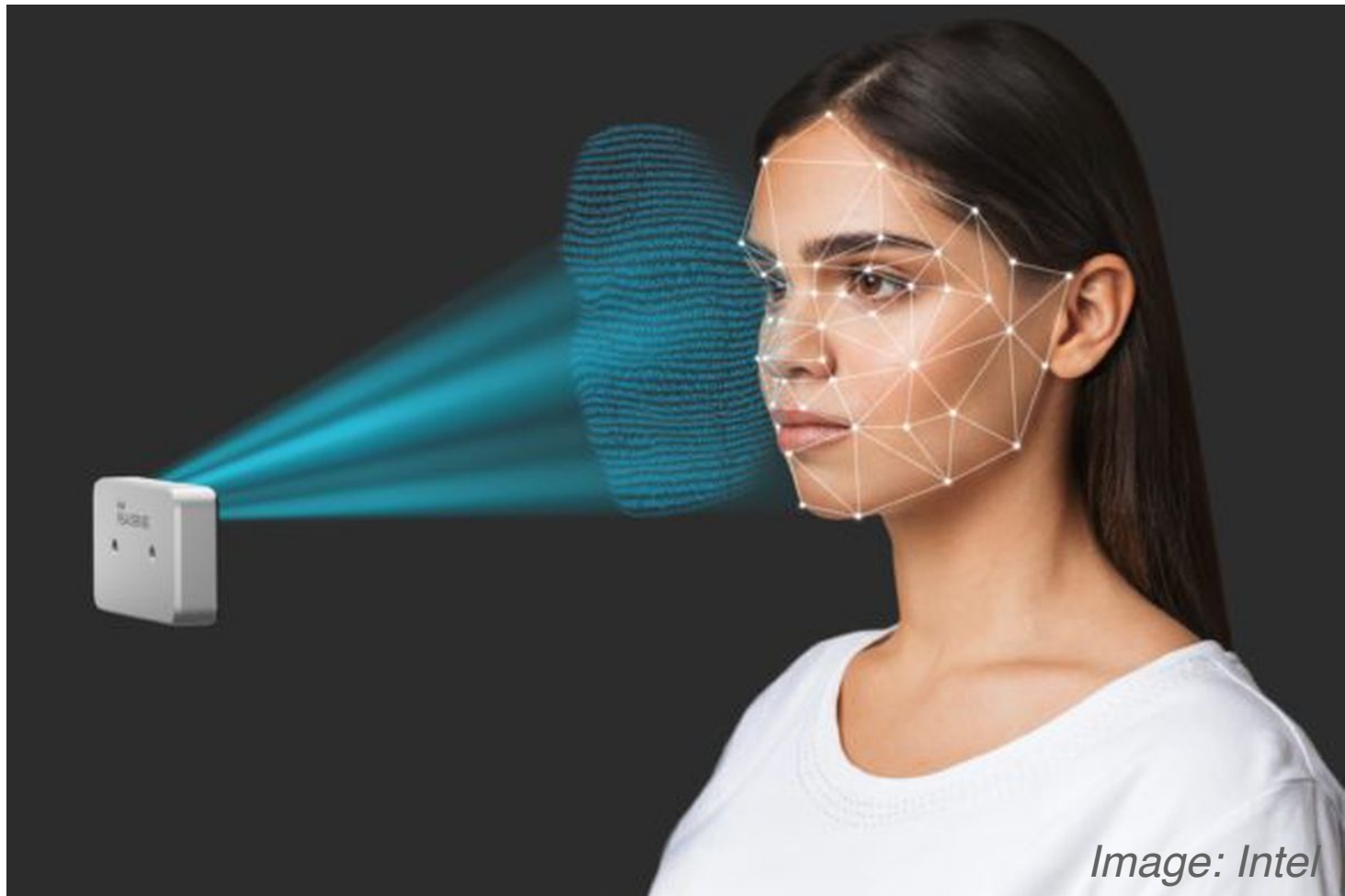
Credit: <https://www.cs.cornell.edu/courses/cs5670/2021sp/>

But humans can tell a lot about a scene from a little information...



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- **What is Computer Vision Used For?**
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Face Reconstruction and Recognition



Face ID



Vending Machine with Face Detection

Disclaimer: I was not part of this research project.
This video contains my commentary on this work.

First Order Motion Model for Image Animation

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Abstract

Image animation consists of generating a video sequence so that an object in a source image is animated according to the motion of a driving video. Our framework addresses this problem without using any annotation or prior information about the specific object to animate. Once trained on a set of videos depicting objects of the same category (e.g. faces, human bodies), our method can be applied to any object of this class. To achieve this, we decouple appearance and motion information using a self-supervised formulation. To support complex motions, we use a representation consisting of a set of learned keypoints along with their local affine transformations. A generator network models occlusions arising during target motions and combines the appearance extracted from the source image and the motion derived from the driving video. Our framework scores best on diverse benchmarks and on a variety of object categories. Our source code is publicly available¹.

1 Introduction

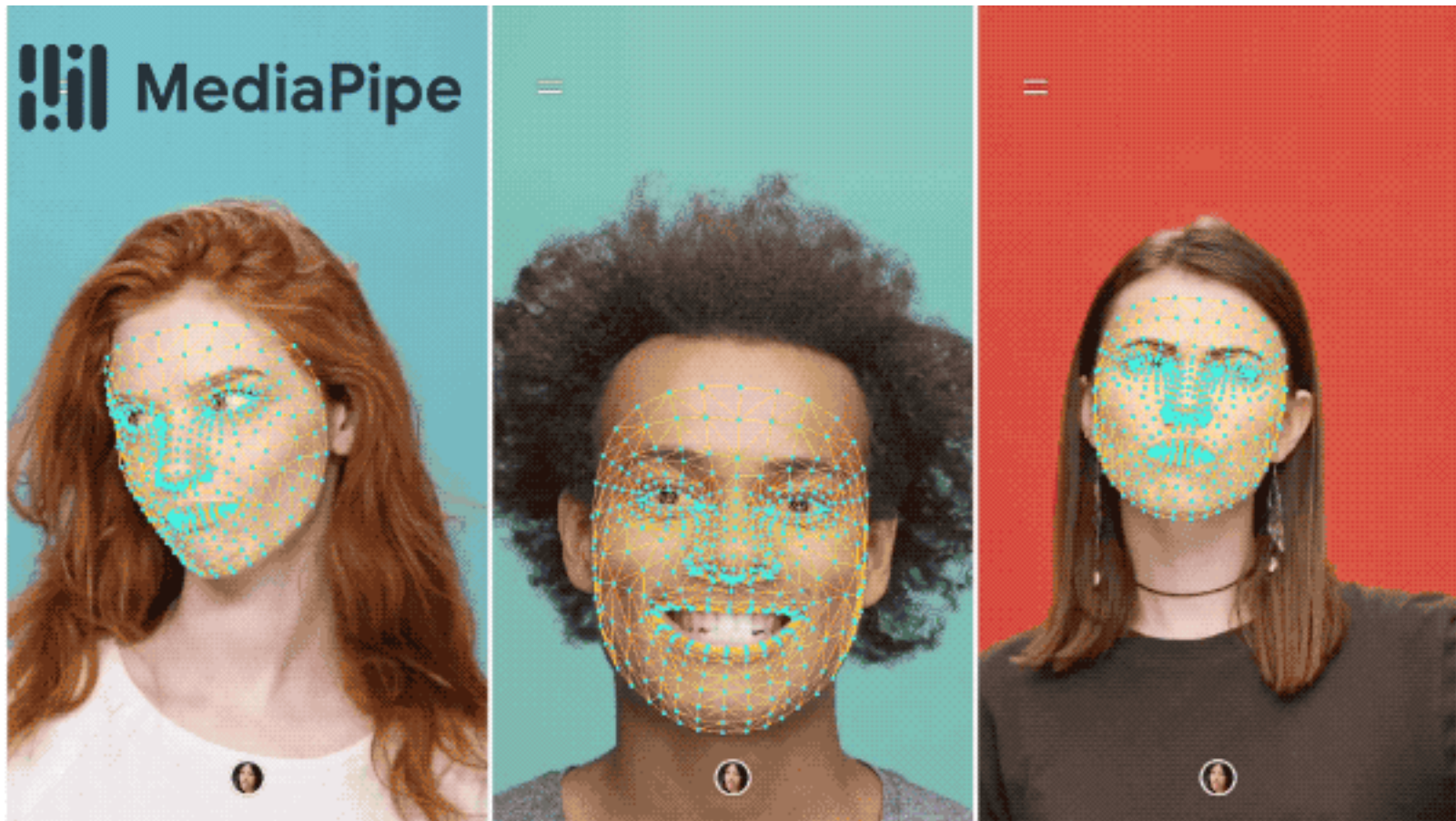
Generating videos by animating objects in still images has countless applications across areas of interest including movie production, photography, and e-commerce. More precisely, image animation refers to the task of automatically synthesizing videos by combining the appearance extracted from a source image with motion patterns derived from a *driving video*. For instance, a face image of a certain person can be animated following the facial expressions of another individual (see Fig. 1). In the literature, most methods tackle this problem by assuming strong priors on the object representation (e.g. 3D model) [4] and resorting to computer graphics techniques [6, 33]. These approaches can be referred to as *object-specific* methods, as they assume knowledge about the model of the specific object to animate.

Recently, deep generative models have emerged as effective techniques for image animation and video retargeting [2, 41, 3, 42, 27, 28, 37, 40, 31, 21]. In particular, Generative Adversarial Networks (GANs) [14] and Variational Auto-Encoders (VAEs) [20] have been used to transfer facial expressions [37] or motion patterns [3] between human subjects in videos. Nevertheless, these approaches usually rely on pre-trained models in order to extract object-specific representations such as keypoint locations. Unfortunately, these pre-trained models are built using costly ground-truth data annotations [2, 27, 31] and are not available in general for an arbitrary object category. To address this issue, recently Siarohin *et al.* [28] introduced Monkey-Net, the first object-agnostic deep model for image

¹<https://github.com/AliaksandrSiarohin/first-order-model>



DeepFake



<https://codepen.io/mediapipe/details/KKgVaPJ>

Augmented reality



Credit: <https://fpcv.cs.columbia.edu/>

Factory Automation



I.1

Credit: <https://fpcv.cs.columbia.edu/>

Vision Inspection



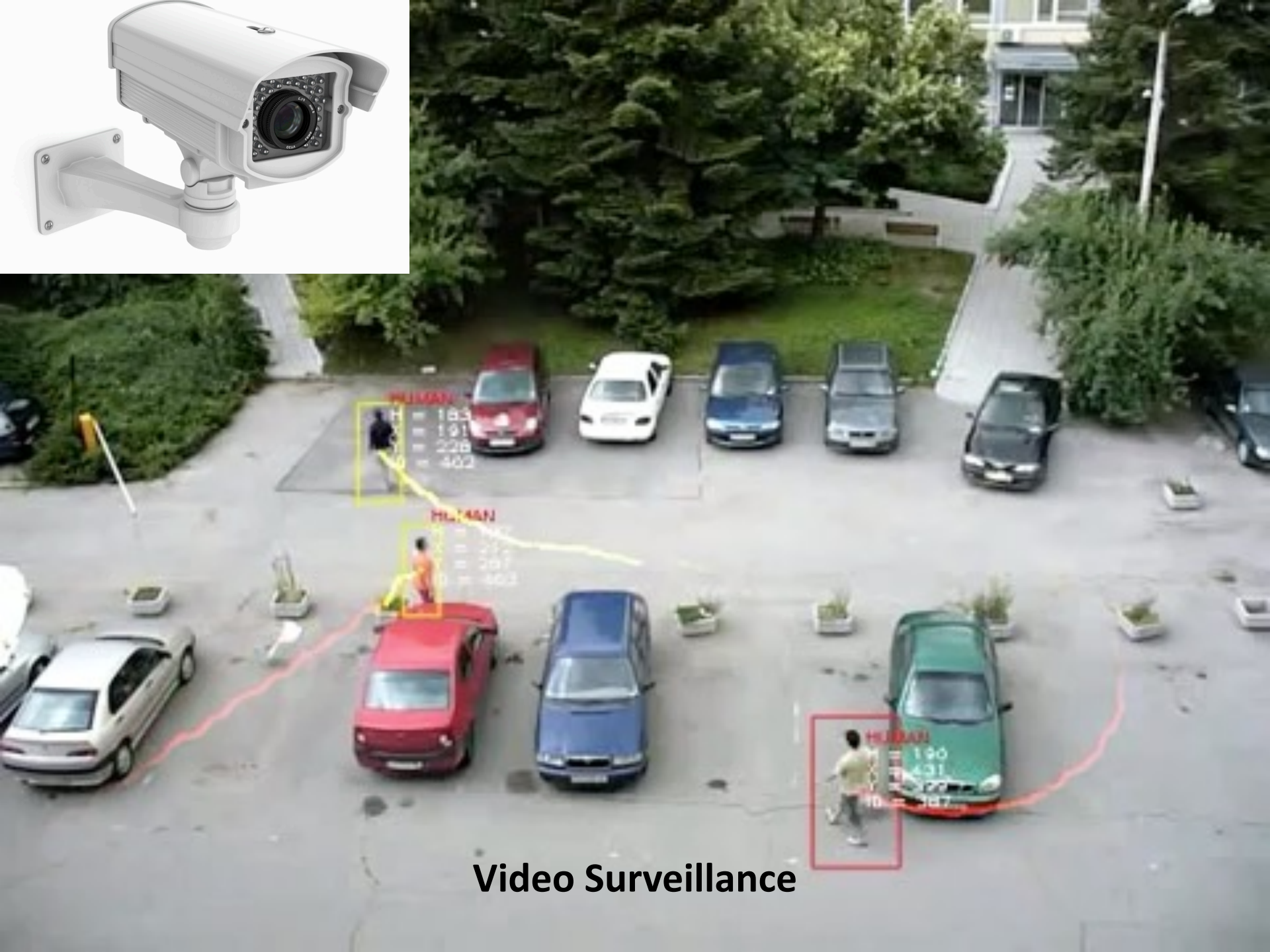
ATA 010

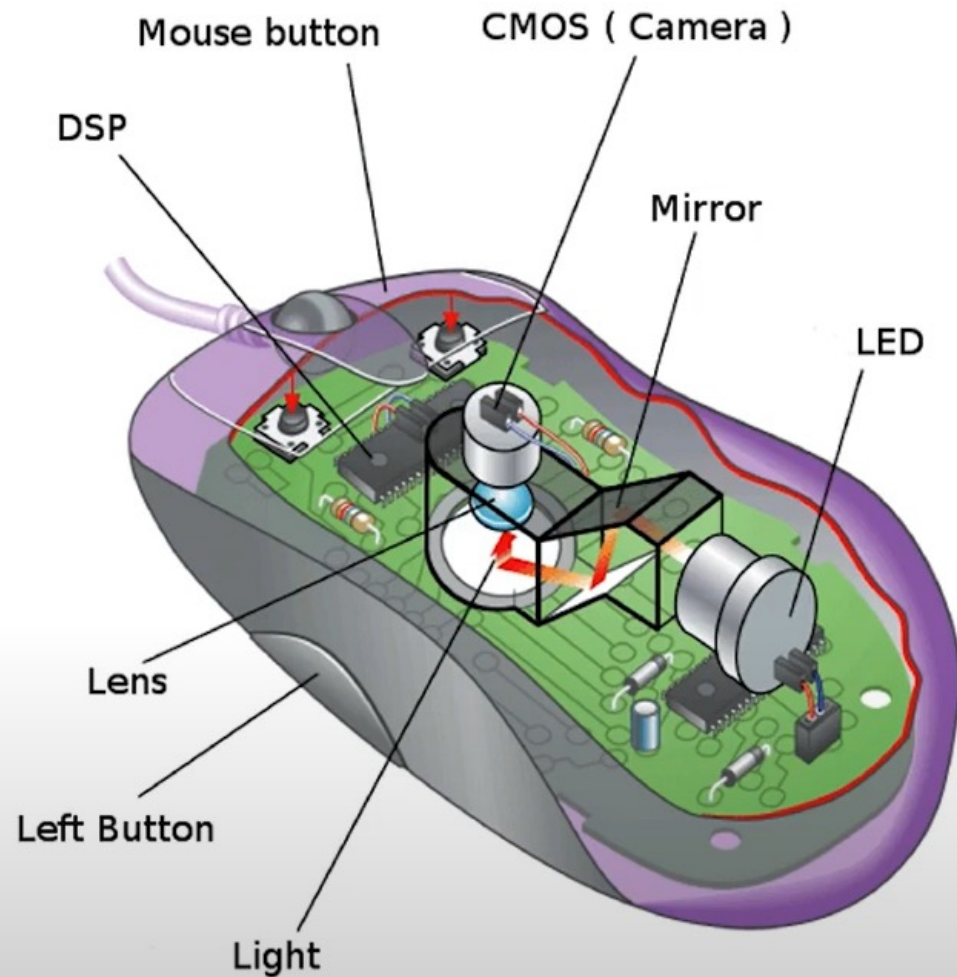
Credit: <https://fpcv.cs.columbia.edu/>

Optical Character Recognition(OCR)



Optical Character Recognition(OCR)





Credit: <https://fpcv.cs.columbia.edu/>

Human Computer Interaction: Optical Mouse



Human Computer Interaction: Motion Sensing Games



KINECT



Human Computer Interaction: Motion Sensing Games

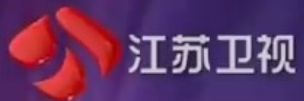


Visual Effects (VFX)



Digital Human

Digital Human

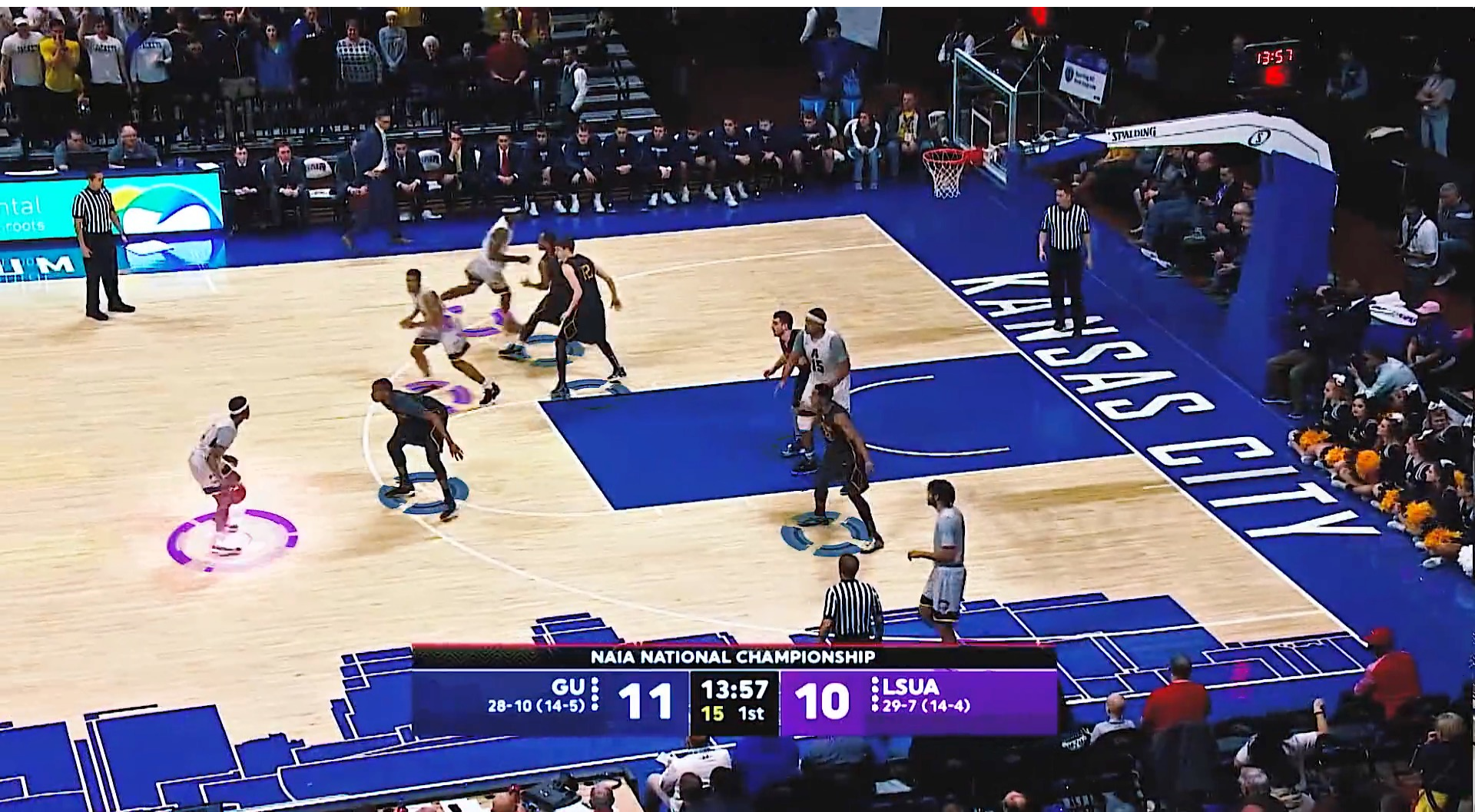


直播
高清
环绕声

周深
收获特别多

元气森林

Digital Human



Sports Broadcasting



[3D Street View](#)



<https://realsee.com/website/product/vr>

VR Tour

室内导航



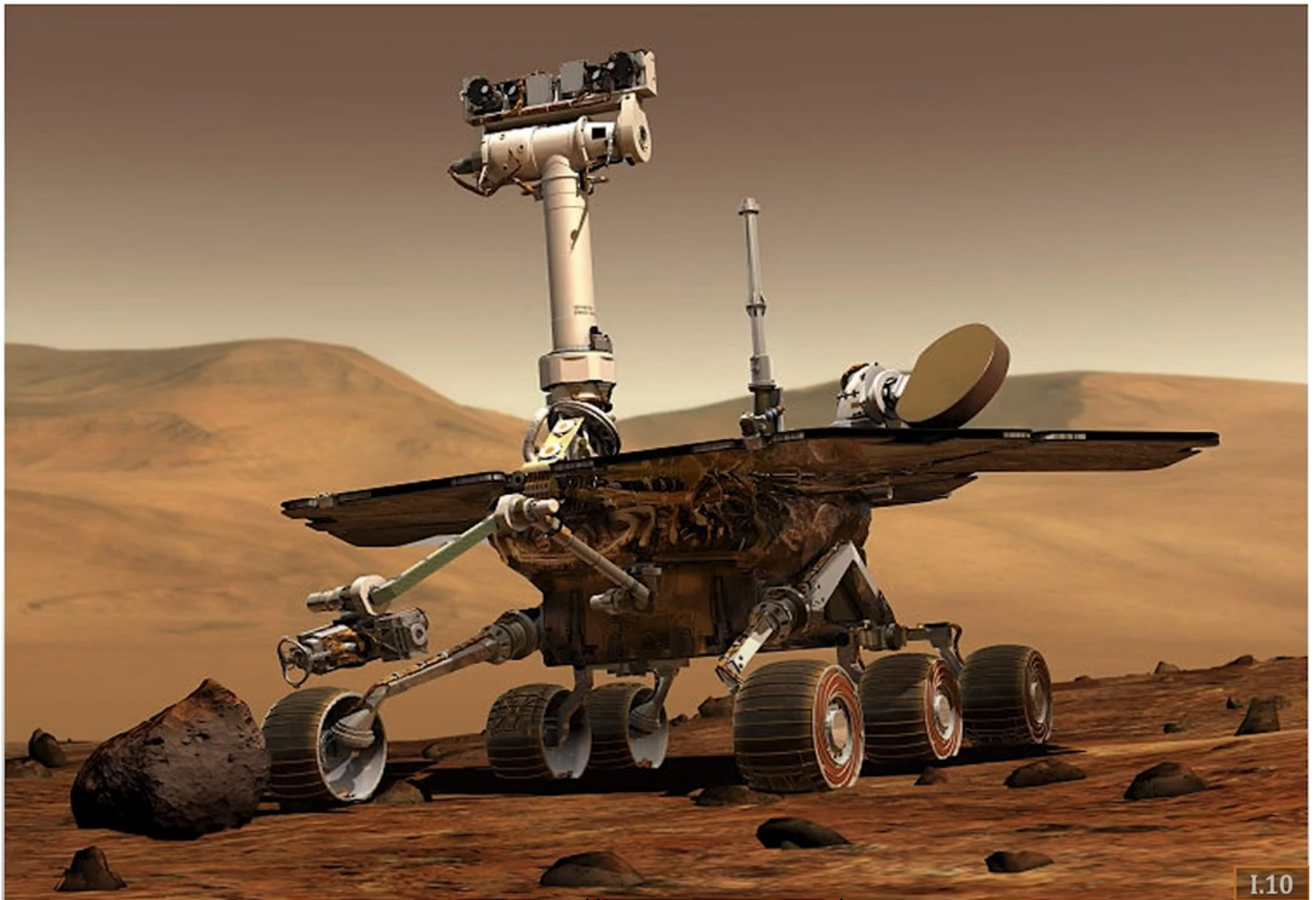
2019
商汤人工智能峰会
SenseTime AI Summit

AI+
大爱无疆

签到处
Reception

请拍摄墙面

Visual Localization and Navigation



Credit: <https://fpcv.cs.columbia.edu/>

Autonomous Navigation: Space Exploration



https://www.youtube.com/watch?v=y3RIHnK0_NE

Robot Perception



Robot Perception

Autonomous Driving



Magic Leap

Augmented Reality

Augmented Reality

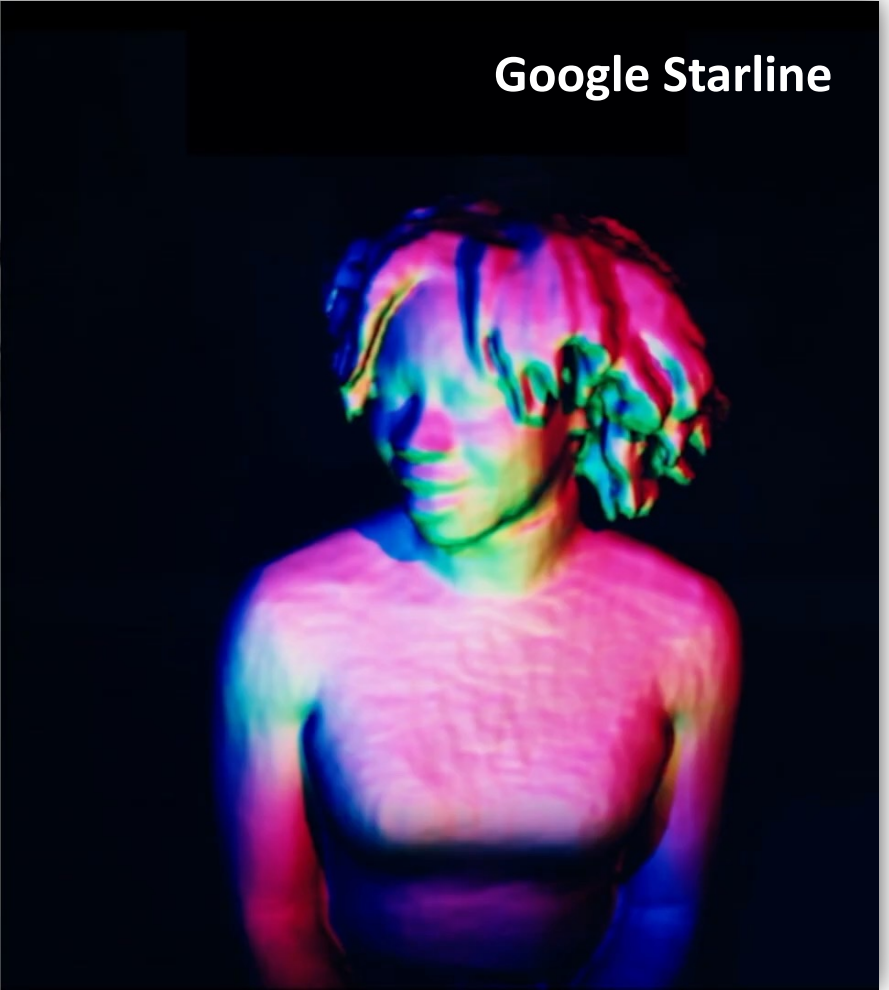


 Microsoft
HoloLens



Microsoft Holoportation

Free Viewpoint Video: Telepresence



<https://www.youtube.com/watch?v=Q13CishCKXY>

Free Viewpoint Video: Telepresence



RED Lion movie shorts

Credit: *The Matrix*

Free Viewpoint Video: Bullet Time Effect

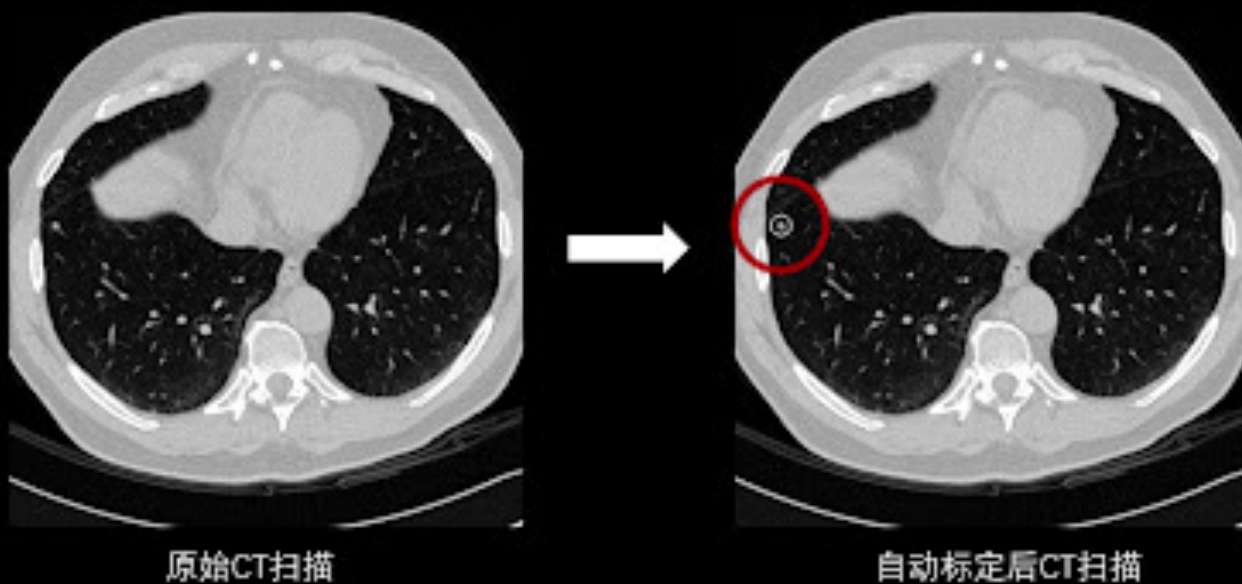


Free Viewpoint Video: Bullet Time Effect



Free Viewpoint Video: Bullet Time Effect

基于CT扫描自动识别恶性结节



Medical Image Analysis



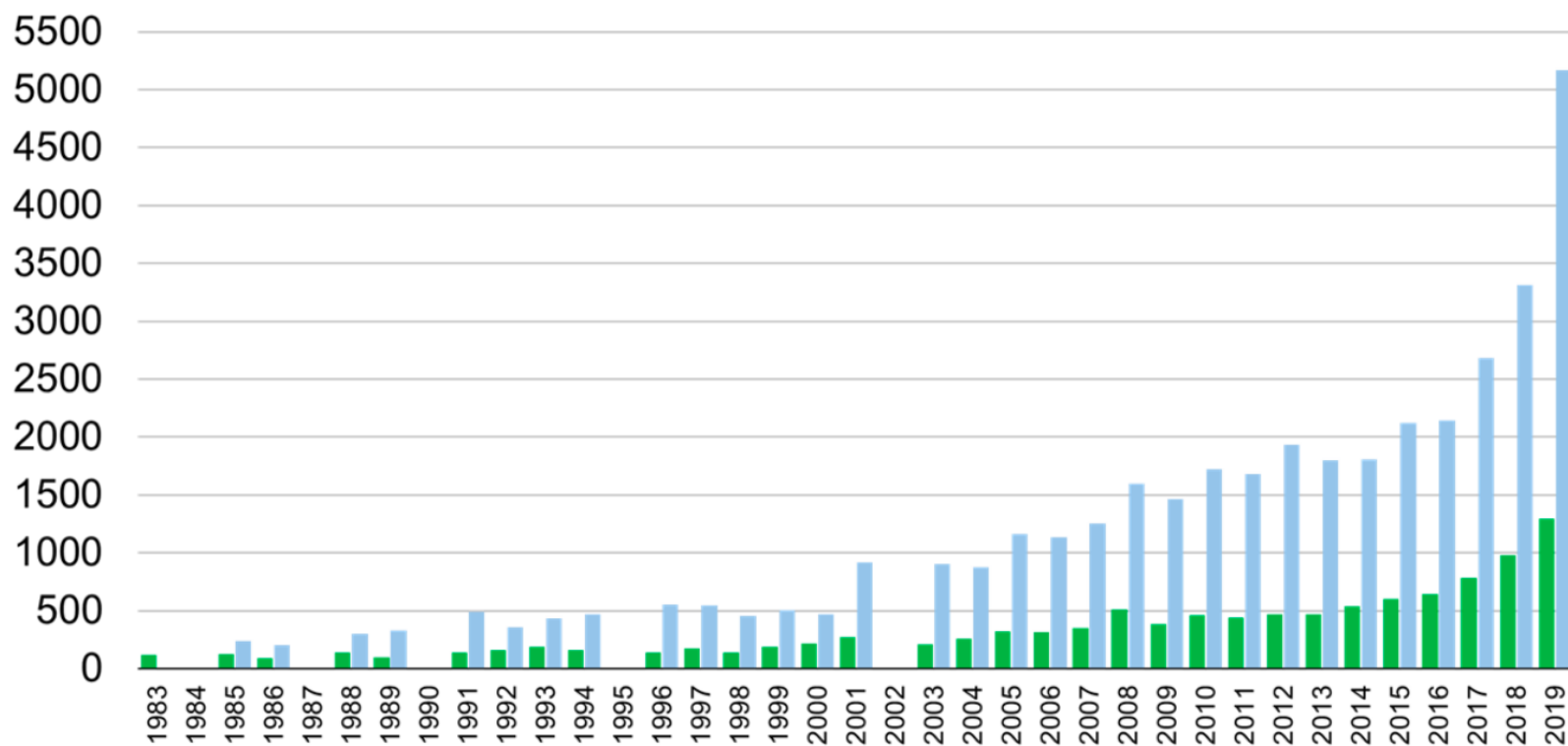
iTero Scanner

Medical Image Analysis

Vision research.

- Vision is a hard problem.
- Vision is Multi-Disciplinary.
- Considerable progress has been made.
- Many successful real-world applications.

Computer vision is an active research area



Number of submitted (blue) and accepted (green) papers in CVPR by year.



ACM SIGGRAPH

Today

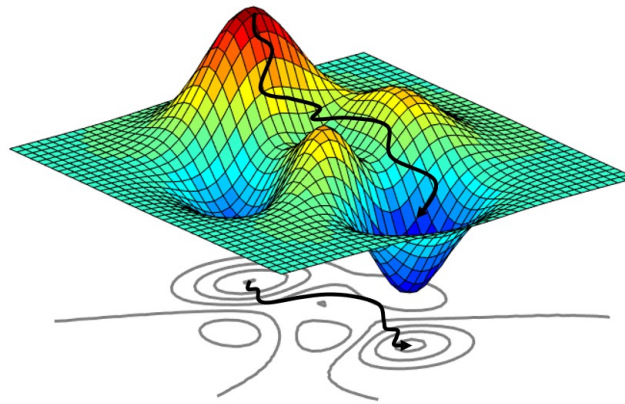
- What is Computer Vision?
- What is Vision Used For?
- **Course Overview.**
- Review of Linear Algebra.

Objective

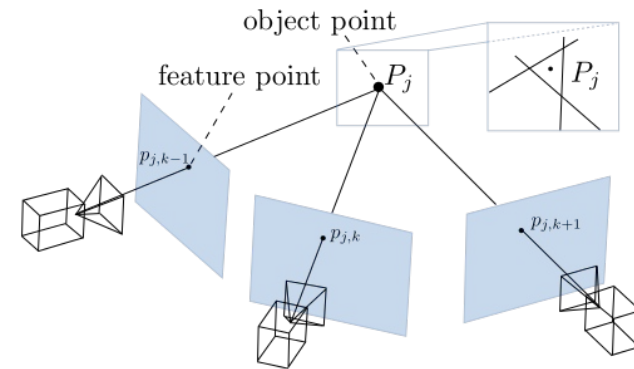
- To be an expert in computer vision
- Methodology: how to formulate and solve problems using mathematics

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Linear algebra



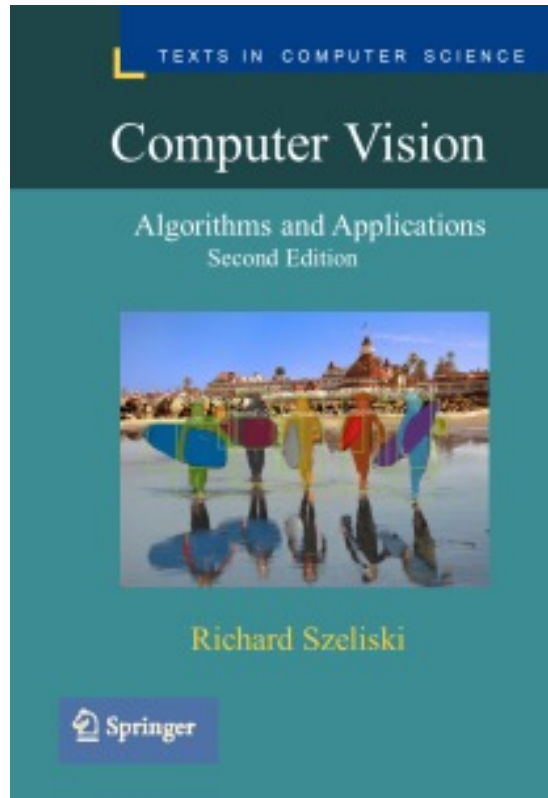
Optimization



Geometry

Textbook

- <http://szeliski.org/Book/>

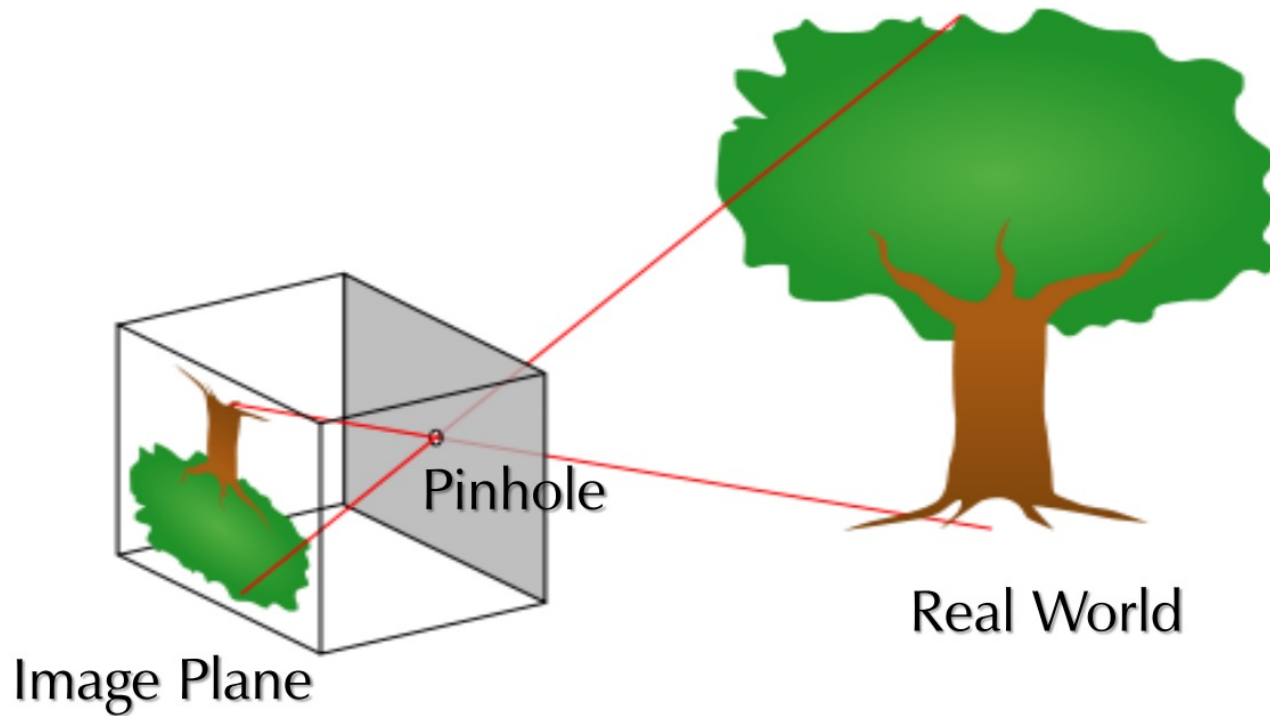


Course overview.

- Basics. (Lec.02 – Lec.04)
- Reconstruction. (Lec.05 – Lec.08)
- Understanding. (Lec.09 – Lec.11)
- Synthesis. (Lec. 12 – Lec.13)

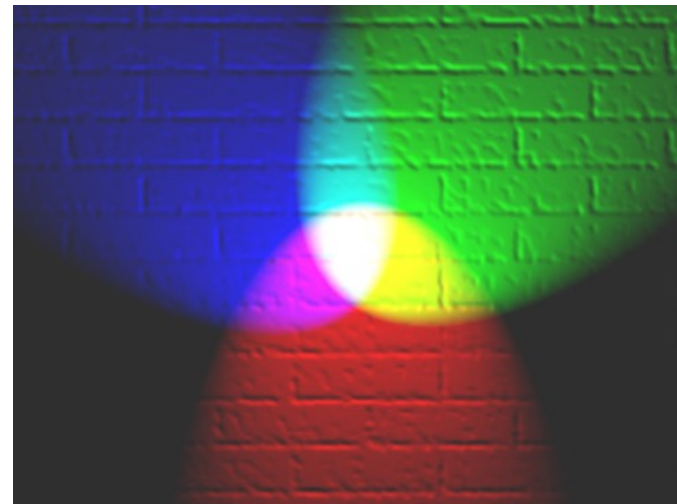
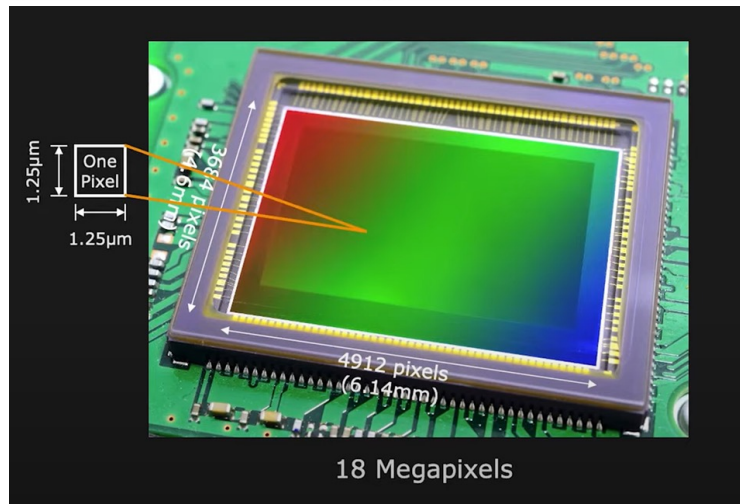
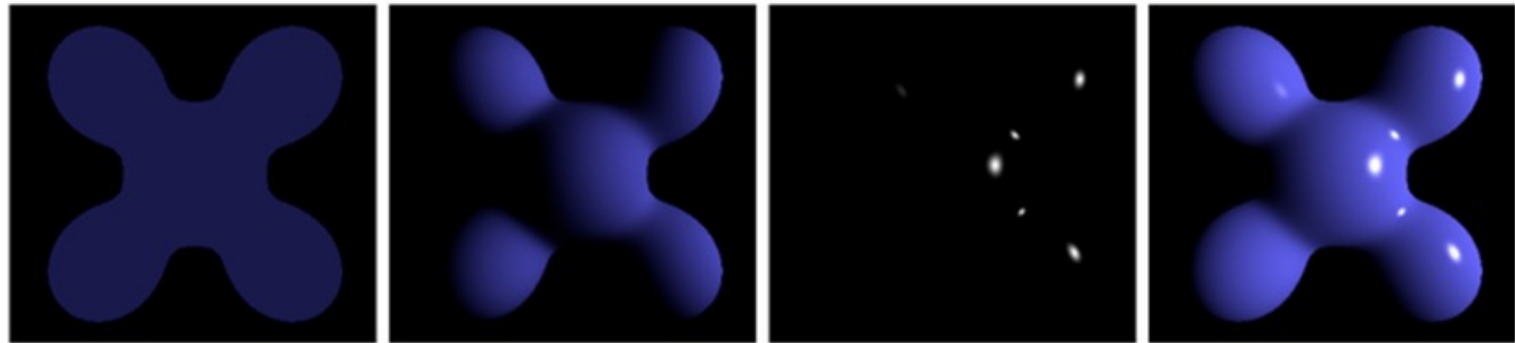
Lec.02: Image formation

- Geometric formation: camera model



Lec.02: Image formation

- Photometric formation: shading, color, sensors



Lec.03: Image processing

- Image filtering



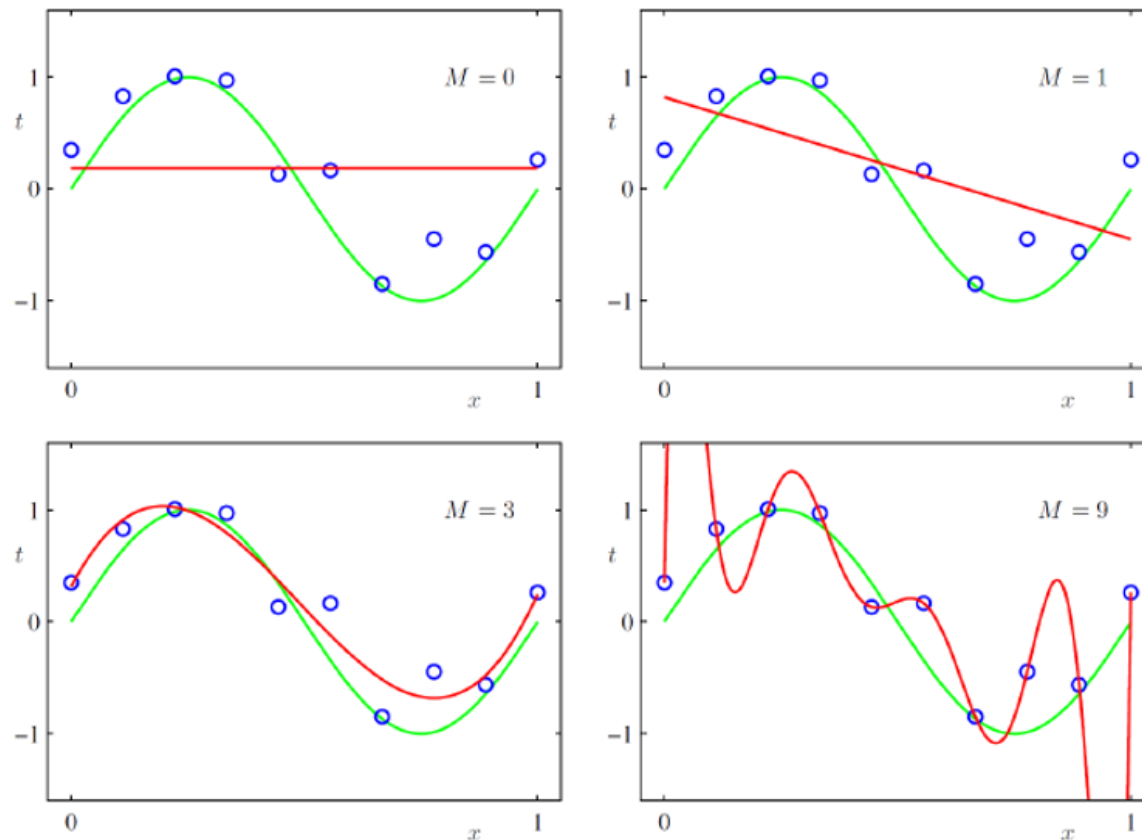
Lec.03: Image processing

- Image resize and reshape



Lec.04: Model fitting and optimization

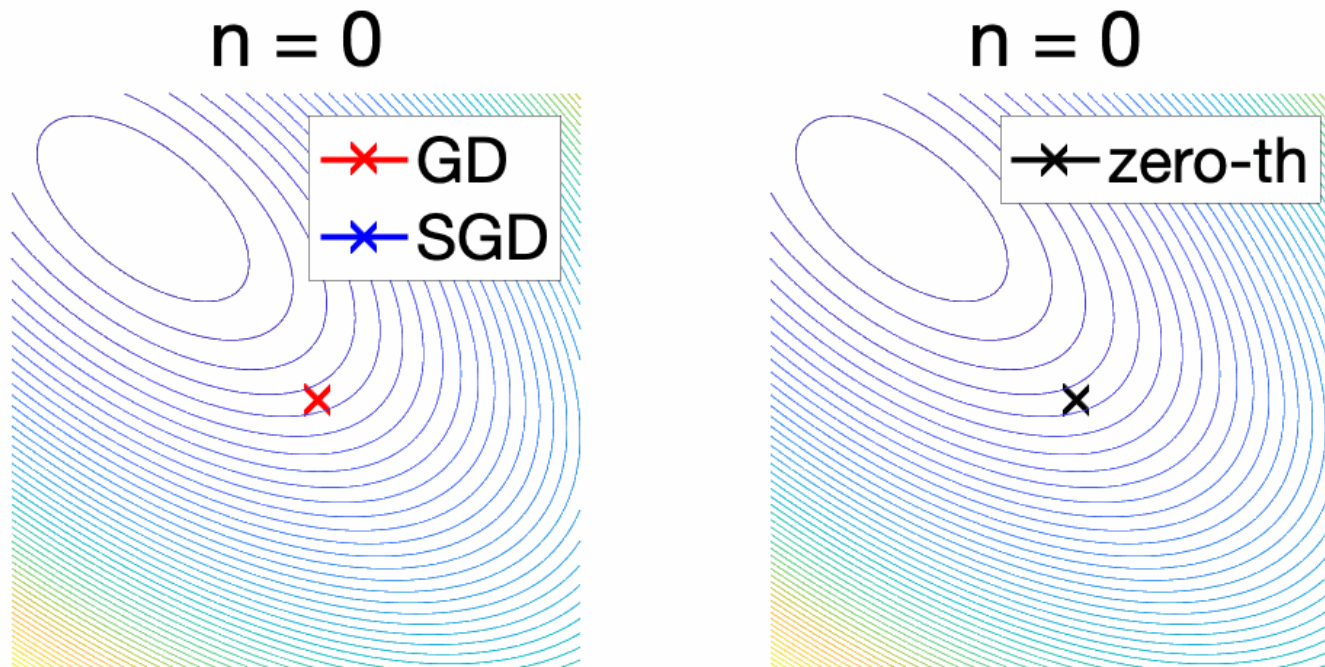
- Model fitting



Credit: <https://szeliski.org/Book/>

Lec.04: Model fitting and optimization

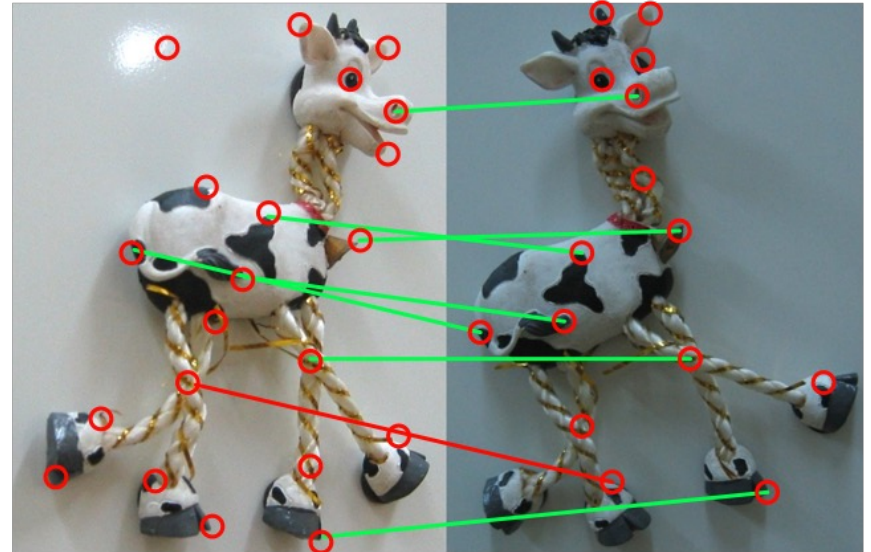
- Mathematical optimization



Credit: <https://francisbach.com/integration-by-parts-randomized-smoothing-score-functions/>
<https://medium.com/@ruhayel/an-intuitive-and-physical-approach-to-newtons-method-86a0bd812ec3>

Lec.05: Feature matching and motion estimation

- Feature matching



Lec.05: Feature matching and motion estimation

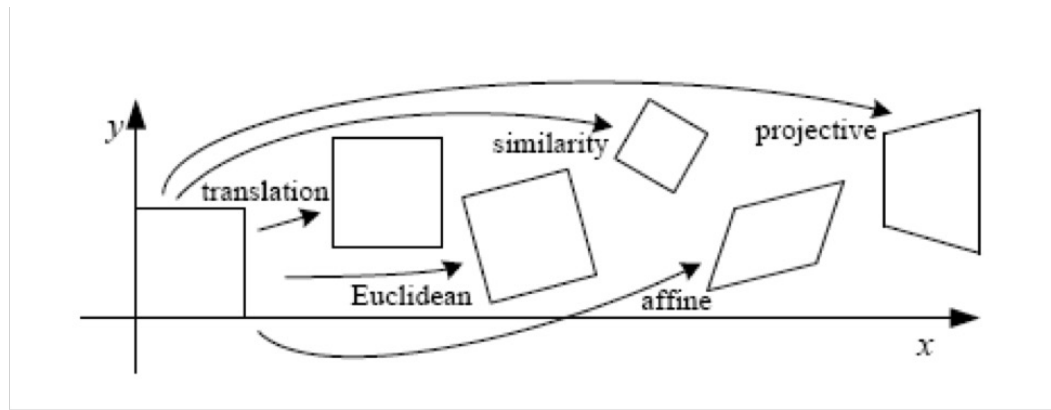
- Motion estimation



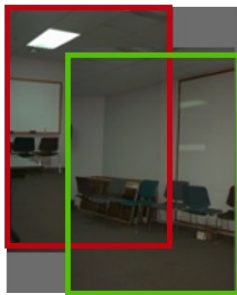
Credit: datahacker.rs/013-optical-flow-using-horn-and-schunck-method/

Lec.06: Image alignment and stitching

- Image transformation

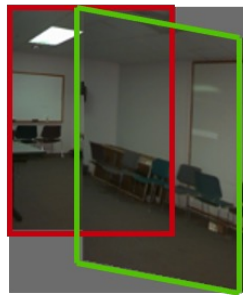


Translation



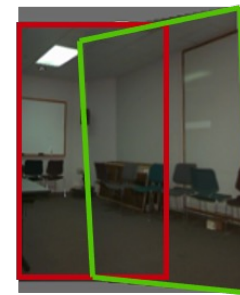
2 unknowns

Affine



6 unknowns

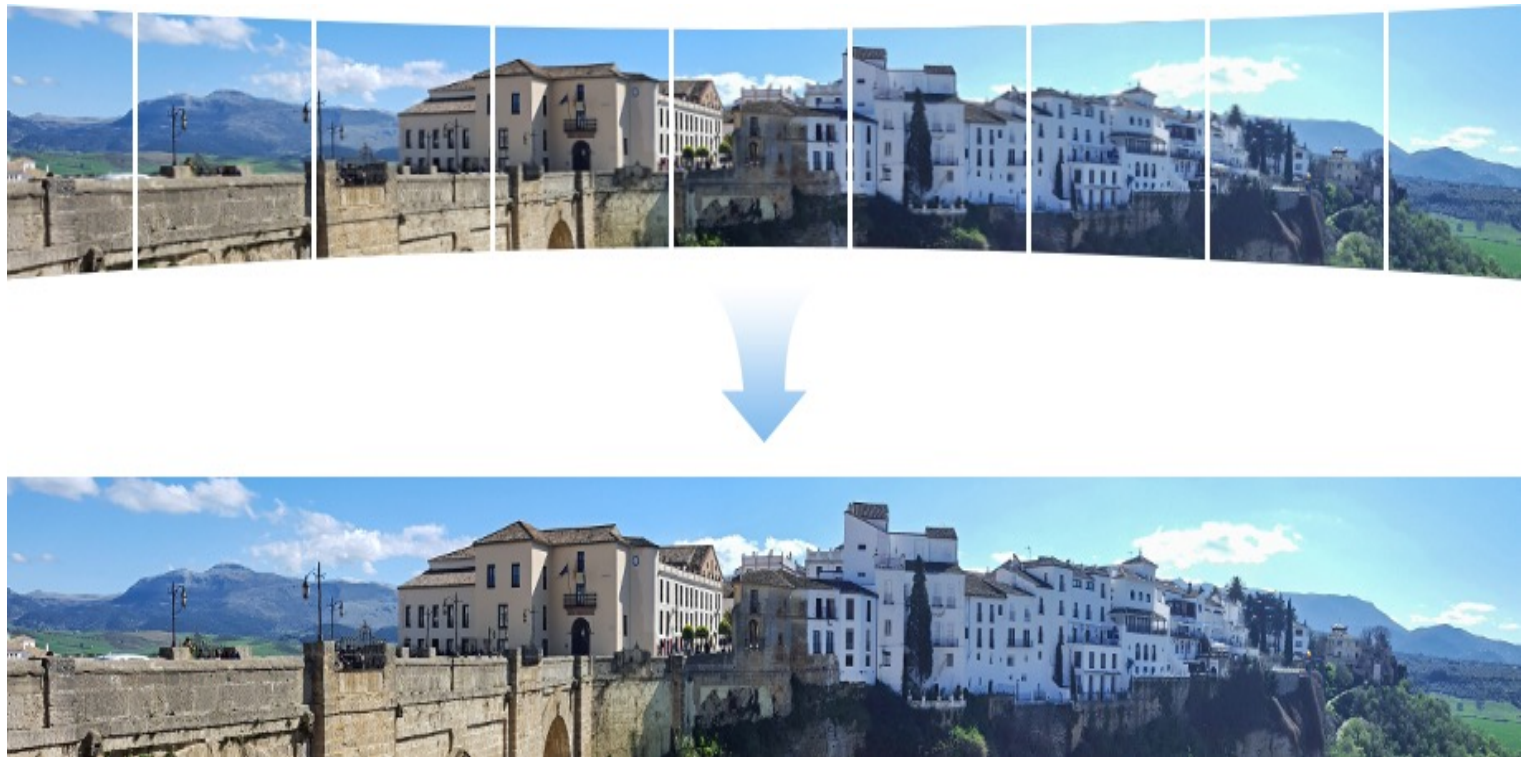
Projective



8 unknowns

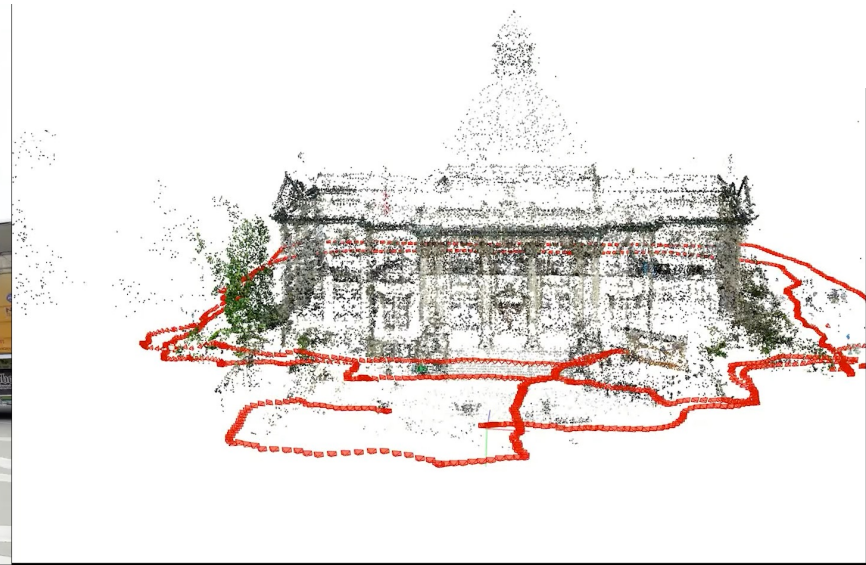
Lec.06: Image alignment and stitching

- Image stitching



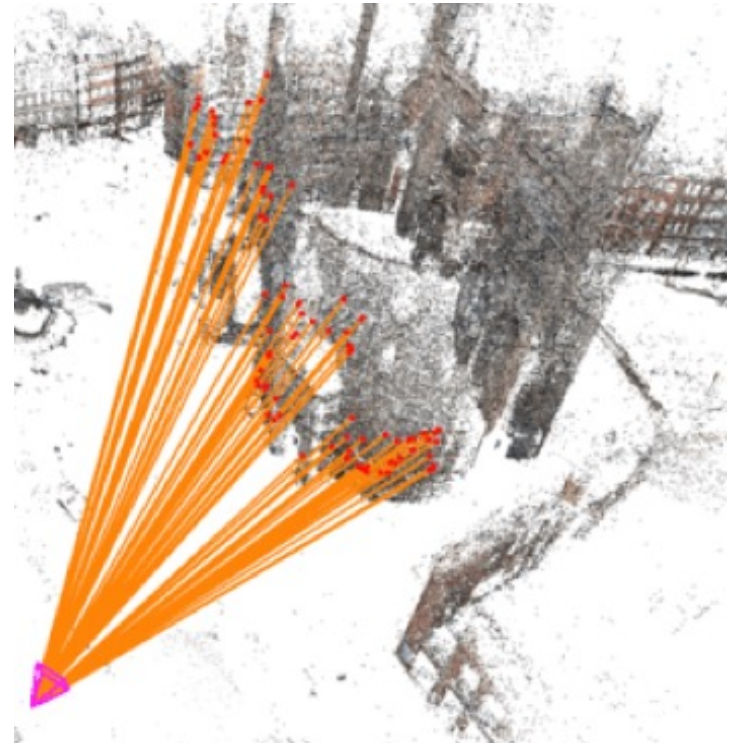
Lec.07: Structure from motion

- Structure from motion



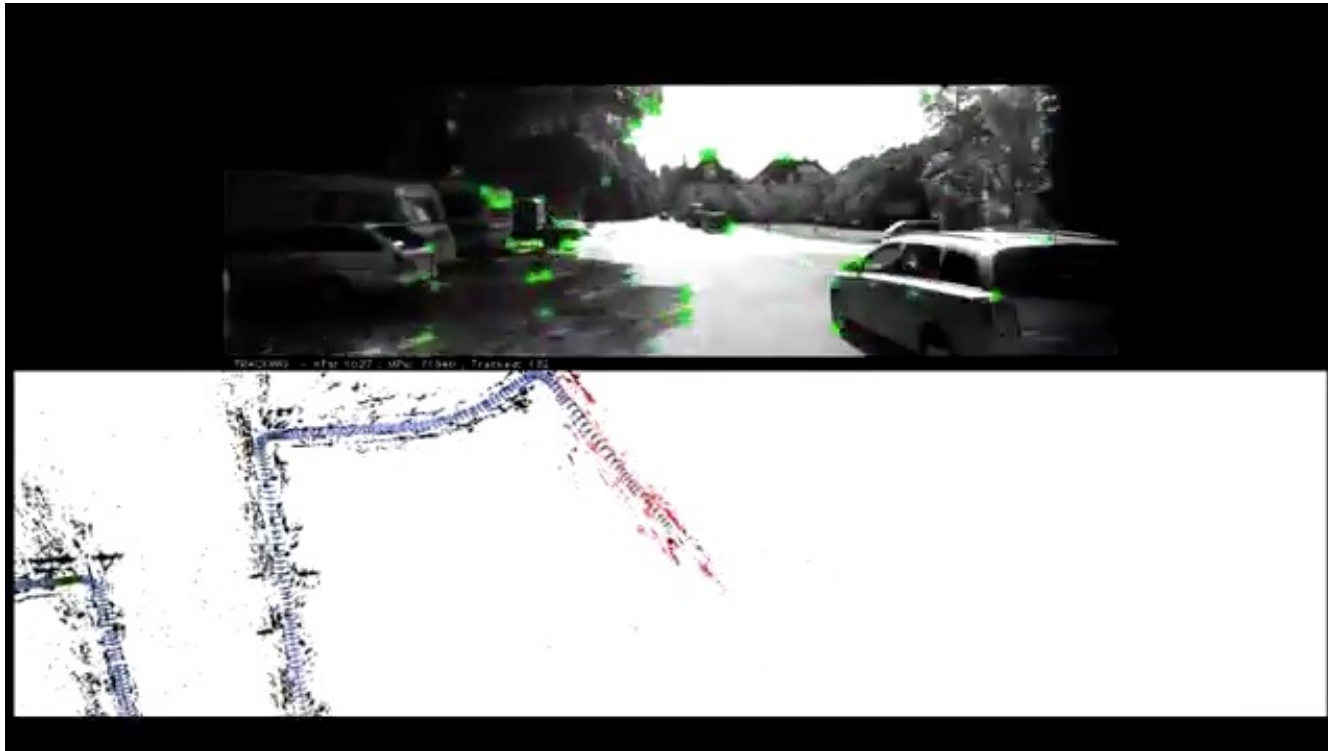
Lec.07: Structure from motion

- Visual localization



Lec.07: Structure from motion

- Simultaneous localization and mapping (SLAM)



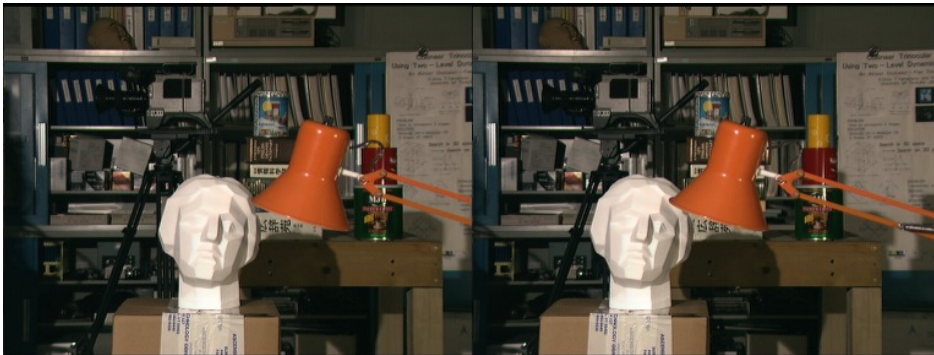
Credit: *ORB-SLAM2, TRO, 2017*

Lec.08: 3D reconstruction



Lec.08: 3D reconstruction

- Stereo matching



Left-right (rectified) stereo pair



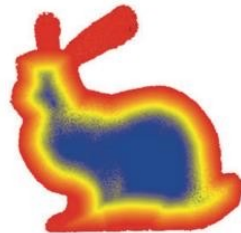
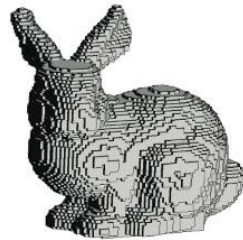
Computed disparity map

Lec.08: 3D reconstruction

- Surface Representations

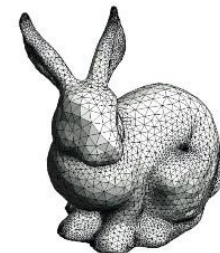
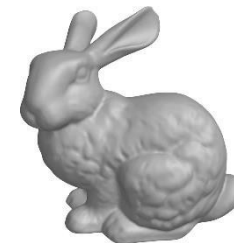
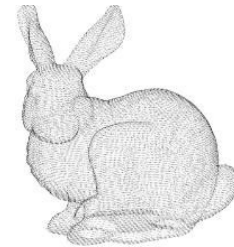
implicit / volumetric

- Voxel grid
- Occupancy grid
- Signed-distance grid



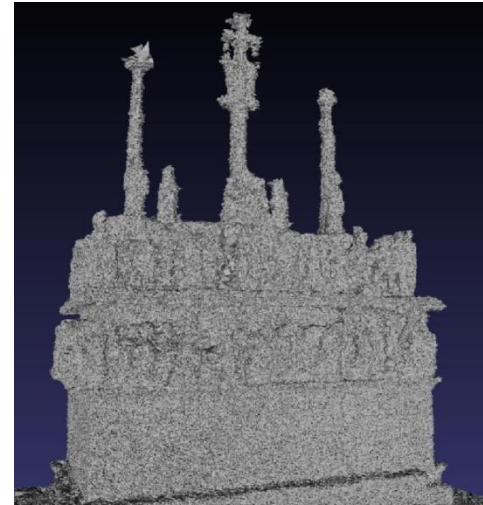
explicit / surface

- Point cloud
- Spline / NURBS
- Surface Mesh

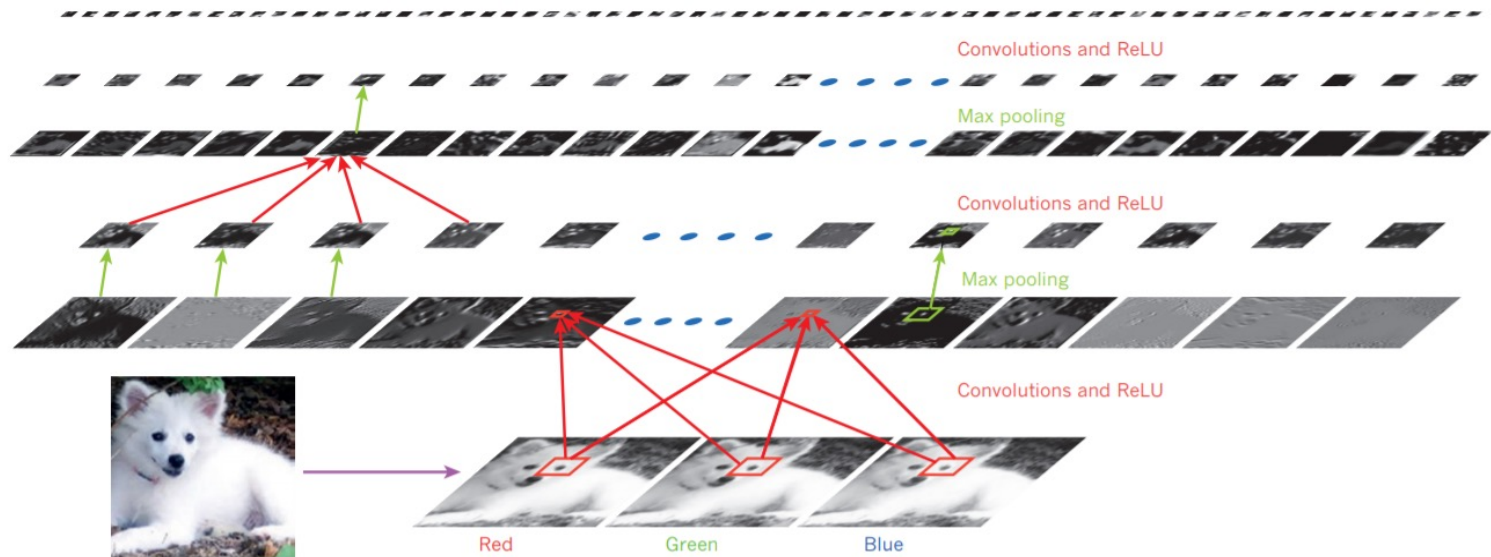


Lec.08: 3D reconstruction

- Surface Extraction from Point Clouds

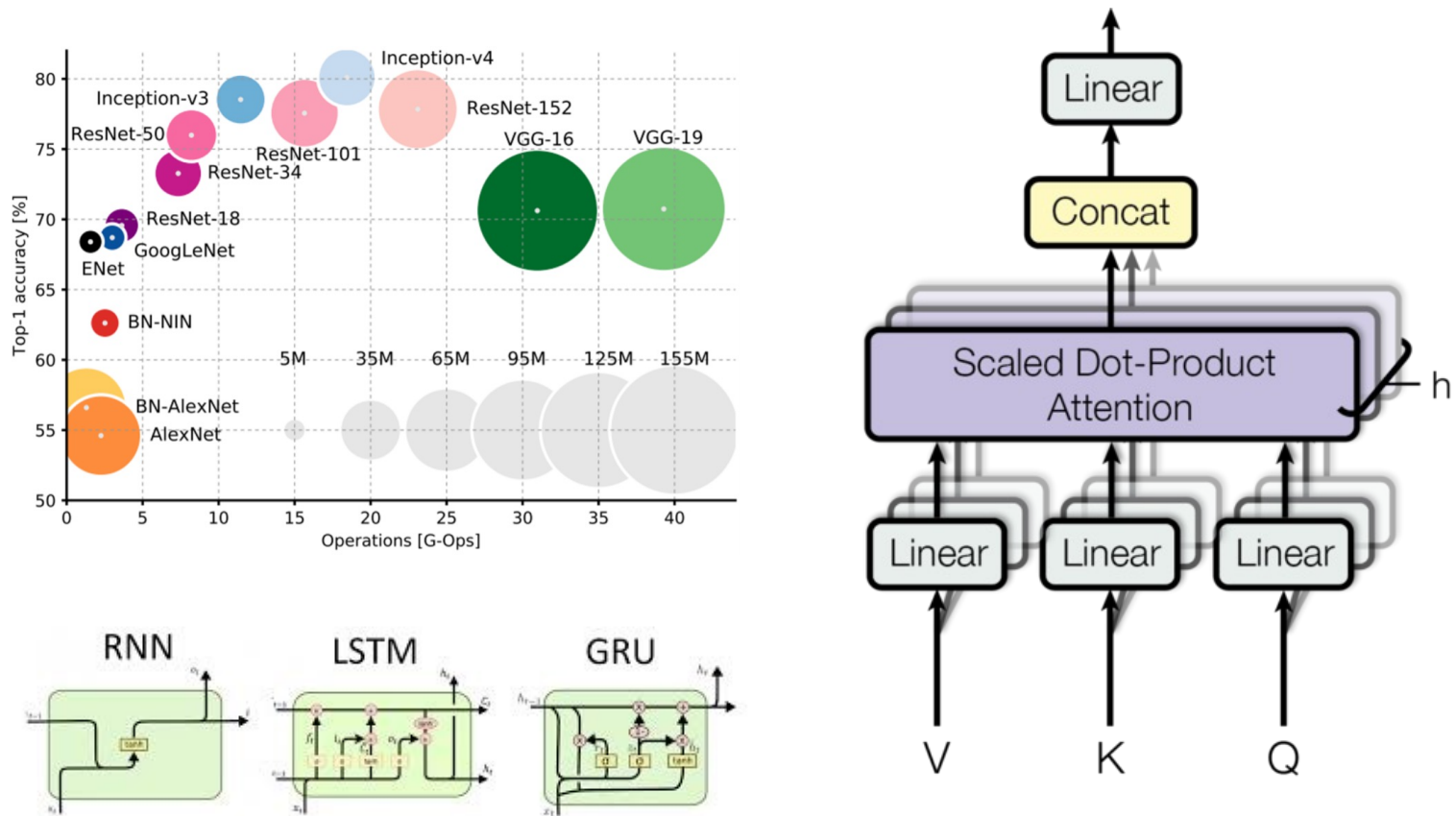


Lec.09: Deep learning



Credit: <https://www.nature.com/articles/nature14539.pdf>

Lec.09: Deep learning



Credit: https://web.eecs.umich.edu/~justincj/slides/eecs498/FA2020/598_FA2020_lecture08.pdf
<https://nlp.seas.harvard.edu/2018/04/03/attention.html>
<https://medium.com/analytics-vidhya/recurrent-neural-network-and-its-variants-de75f9ee063>

Lec.10: Recognition

- Detection and segmentation

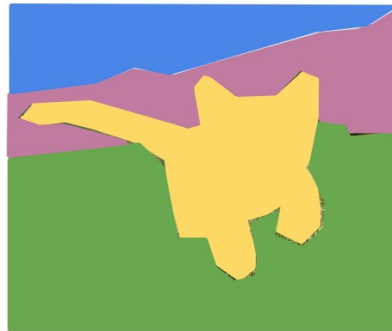
Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Objects

Instance Segmentation



DOG, DOG, CAT

Credit: https://web.eecs.umich.edu/~justincj/slides/eecs498/FA2020/598_FA2020_lecture15.pdf

Lec.10: Recognition

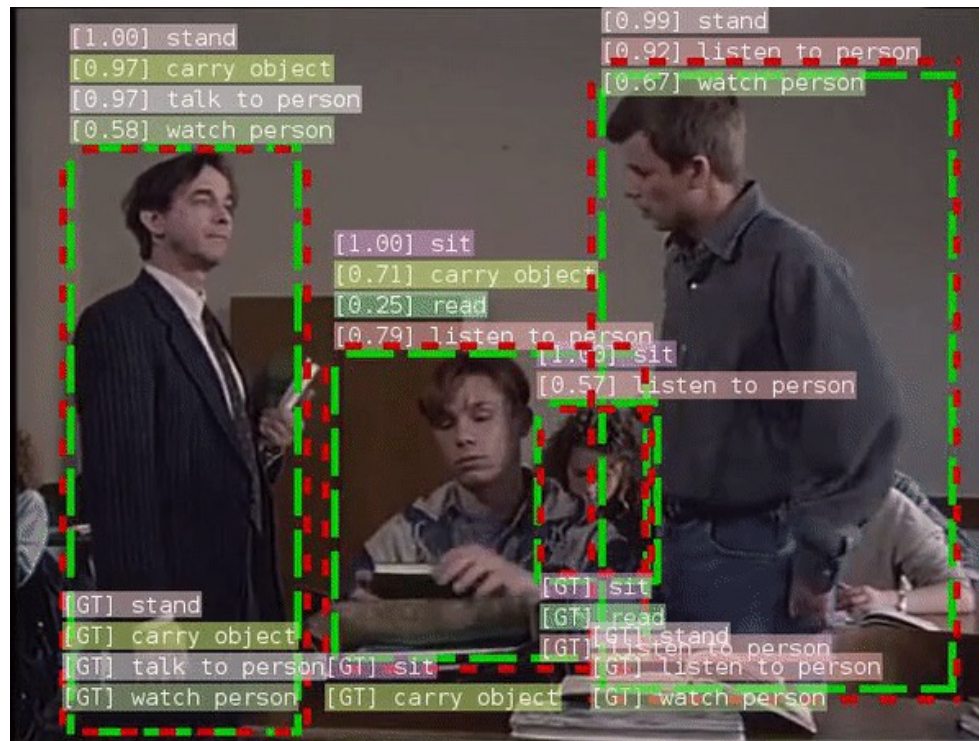
- Human pose estimation



<https://codepen.io/mediapipe/pen/LYRRYEw>

Lec.10: Recognition

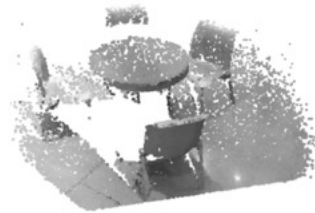
- Action recognition



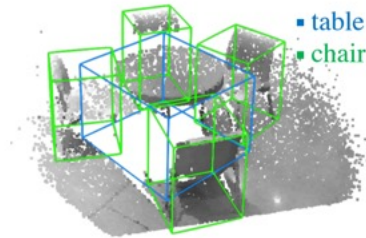
Credit: <https://medium.com/@thomaschaton/slowfast-networks-for-video-recognition-29-oct-2019-iccv-799eb6408d34>

Lec.11: 3D deep learning

Input



Output



Credit: https://cseweb.ucsd.edu/~haosu/slides/3ddl_tutorial_version_20200403.pdf

Lec.12: Computational photography I

- High dynamic range (HDR)



Lec.12: Computational photography I

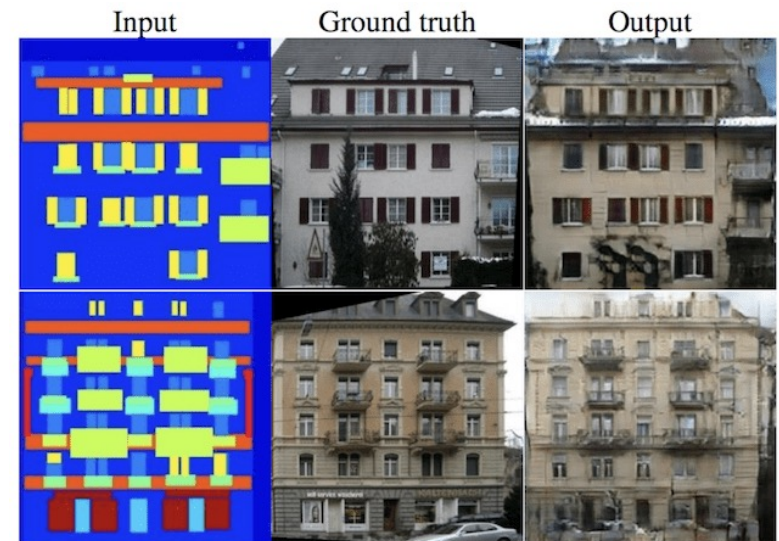
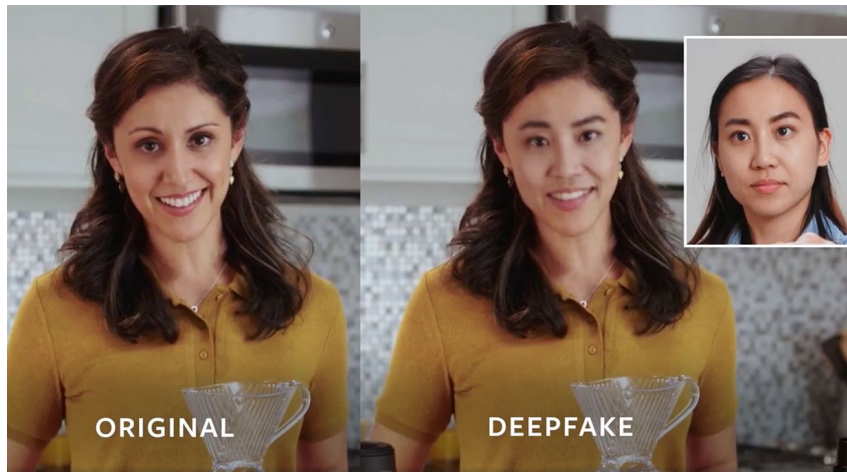
- Super-resolution



Credit: <https://community.arm.com/developer/ip-products/processors/b/ai-and-ml-blog/posts/super-resolution-on-arm-npu>

Lec.12: Computational photography I

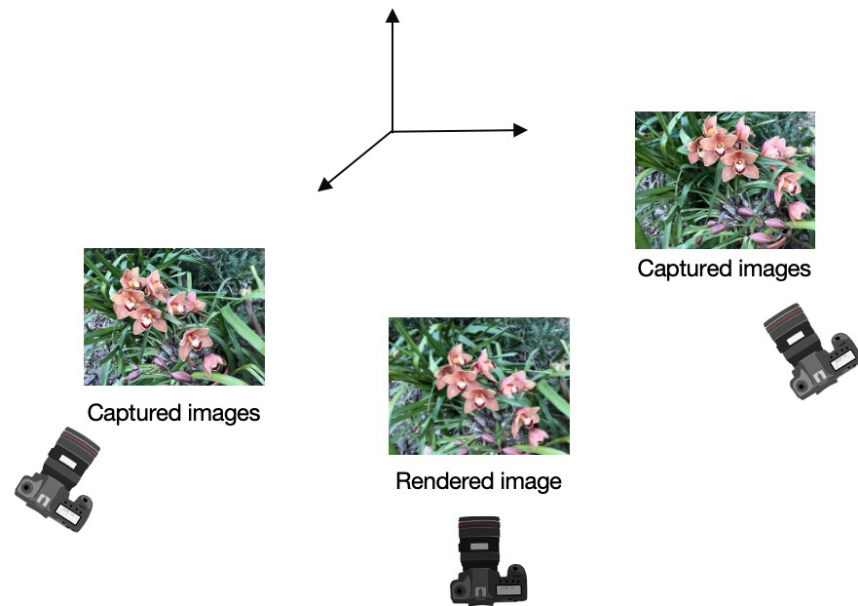
- Image-to-image translation



Credit: Facebook Deepfake detection challenge
\Image-to-Image Translation With Conditional Adversarial Networks.

Lec.13 Computational photography 2

- Neural rendering



Lec.14&15: Final project presentation



© CanStockPhoto.com

考核方式

- 平时作业： 30%
- 大作业： 40%
- 期末考试： 30%

实验课

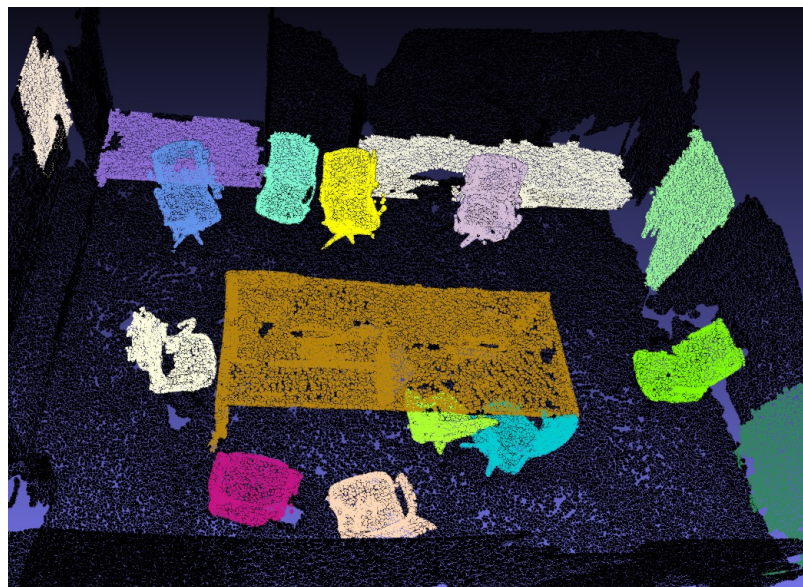
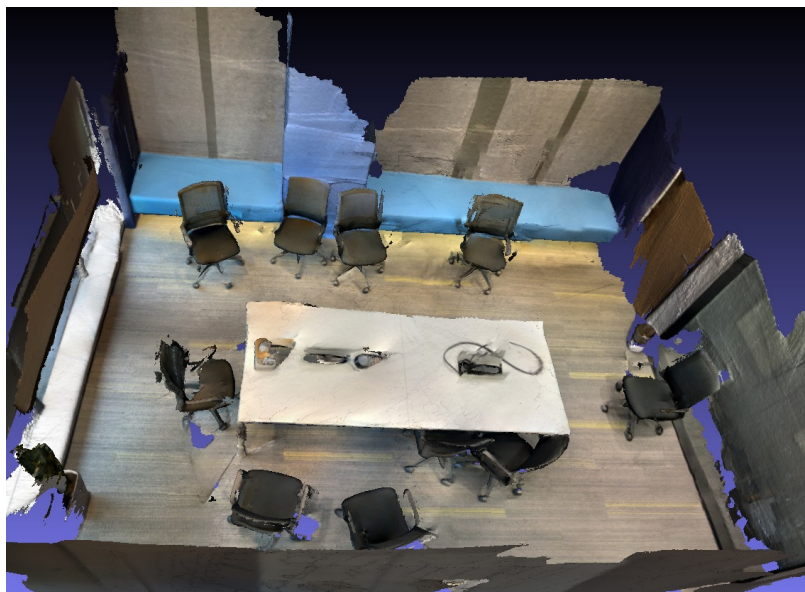
- 时间：单周周二上午1-2节
- 地点：玉泉曹光彪西-503
- 内容：教你动手写代码
- 作业：实验课布置，可在实验课内完成

平时作业

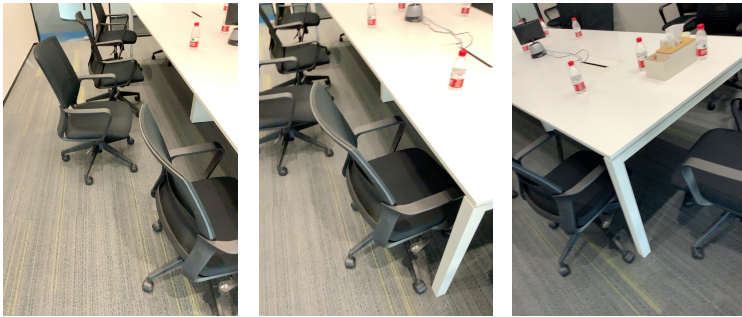
- 每次实验课后布置，共6次
- 通过动手编程来了解概念与算法
- 都对大作业有帮助

去年的大作业

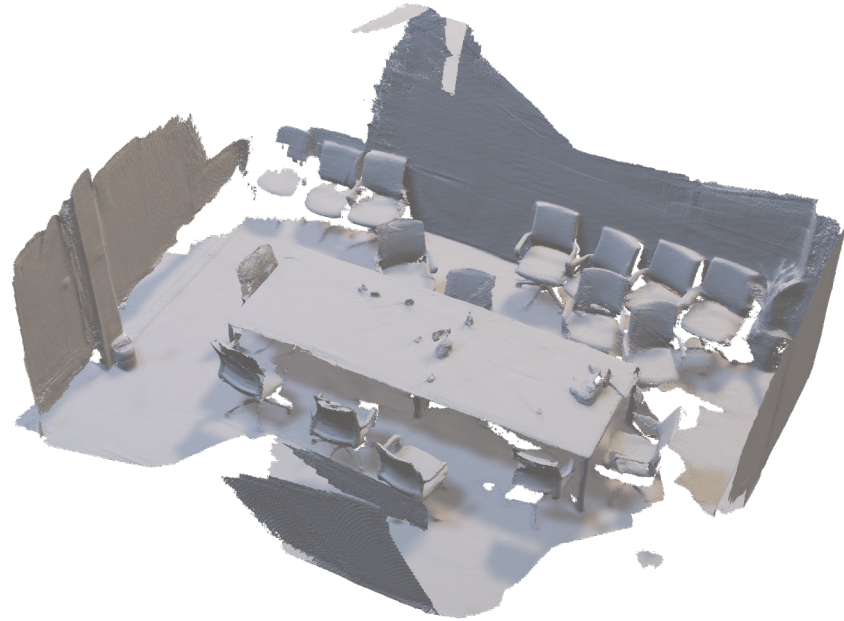
三维场景重建和语义理解



Task 1: 3D Reconstruction



Input Video

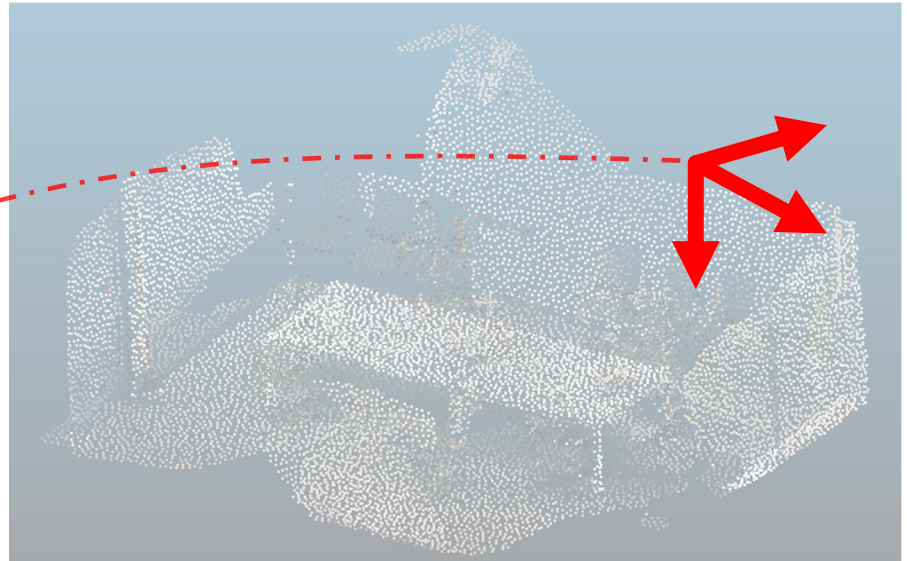


3D model

Task 2: Visual Localization

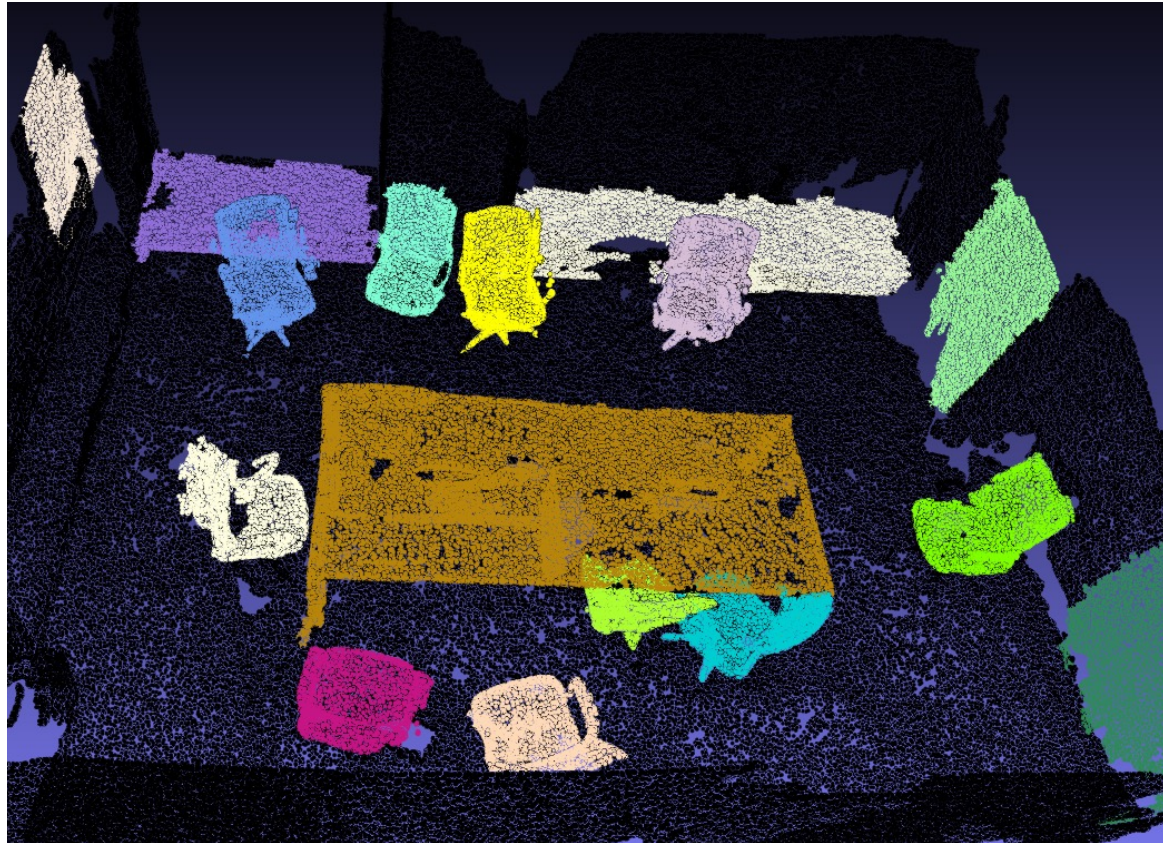


Query Image



Camera Position and Orientation

Task 3: Segmentation



大作业

- 最多两人一组
- 提供基础框架
- 提供GPU服务器
- 评分标准
 - 完成项目可得基础分
 - 定量评价：重建精度、定位精度、分割精度
 - 对以上某方面进行研究，尝试改进
 - 加分：实现某种应用

本次课的额外作业

- 学习PPT里的英文术语